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Jens H. E. Christensen
Federal Reserve Bank of San Francisco

Sarah Mouabbi
Banque de France

Caroline Paulson
Federal Reserve Bank of San Francisco

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German Inflation-Linked Bonds: Overpriced, yet Undervalued

Jens H. E. Christensen[†]

Sarah Mouabbi[‡]

Caroline Paulson^{*}

Abstract

We document that German inflation-linked government bond yields contain a convenience or safety premium averaging 0.33 percent. Despite that, the German Federal Finance Agency decided to cease all future issuance of these bonds in November 2023. We examine the market response to this announcement and find that neither the safety premia nor the trading conditions of these bonds have been negatively impacted. Hence, this bond market remains a rich source of information on real rates in the euro area in addition to offering investors a safe inflation-protected asset.

JEL Classification: C32, E43, E52, G12

Keywords: affine arbitrage-free term structure model, financial market frictions, convenience premium, safety premium

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[†]Corresponding author: Federal Reserve Bank of San Francisco, 101 Market Street MS 1130, San Francisco, CA 94105, USA; phone: 1-415-974-3115; e-mail: jens.christensen@sf.frb.org.

[‡]Banque de France; e-mail: sarah.mouabbi@banque-france.fr.

^{*}E-mail: cpaulson75@gmail.com.

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1 Introduction

Standard fixed-coupon German government bonds, widely known as *bunds*, represent the benchmark class of safe assets in the euro area and are second only to U.S. Treasuries in terms of investor base, liquidity, and market depth. In contrast, German inflation-linked government bonds, though important, have received much less attention. This sense of underappreciation was further reinforced by the German Federal Finance Agency’s decision, in November 2023, to cease all future inflation-linked bond issuance. This paper is among the first to provide a comprehensive analysis of the market for German inflation-linked government bonds and to study its response to this extraordinary announcement.

Given the large and well-documented flight-to-safety effects in the German bund market,¹ we conjecture that investors might also be willing to pay a premium, albeit somewhat smaller, for safely storing their wealth in German inflation-linked government bonds despite their lack of market attention. Hence, the main purpose of our analysis is to examine whether there are any convenience premia embedded in the prices of German inflation-linked government bonds, as little is known about the pricing in this market.

To estimate bond-specific convenience premia, we use an arbitrage-free dynamic term structure model of real yields augmented with a bond-specific risk factor. The identification of the bond-specific risk factor comes from its unique loading for each individual bond security as in Andreasen et al. (2021, henceforth ACR). Our analysis uses prices of individual bonds rather than the more common input of yields from fitted synthetic curves. The underlying mechanism assumes that, over time, an increasing proportion of the outstanding inventory is locked up in buy-and-hold investors’ portfolios. Given forward-looking investor behavior, this lock-up effect means that a particular bond’s sensitivity to the market-wide bond-specific risk factor will vary depending on how seasoned the bond is and how close to maturity it is. In a careful study of nominal U.S. Treasuries, Fontaine and Garcia (2012) find a pervasive bond-specific factor that affects all bond prices, with loadings that vary with the maturity and age of each bond. Given that time since issuance and remaining time to maturity are unique to each bond, this feature of the data allows us to identify the overall bond-specific risk factor and each bond’s loading on that factor by observing the cross section of bond prices over time. Moreover, we stress that this technique is particularly useful for analyzing inflation-linked debt when only a limited sample of bonds may be available, as in our case.²

Using this modeling tool, we are the first to document the existence of large and time-varying convenience premia in this market that average 0.33 percent for our sample period from October 2007 to December 2024. Given that the German inflation-linked bonds are

¹See Paret and Weber (2019) for an example.

²Finlay and Wende (2012) examine prices from a limited number of Australian inflation-linked bonds but do not account for bond-specific liquidity or convenience premia.

much less liquid than standard bunds in terms of bid-ask spreads, we follow Christensen and Mirkov (2022) and refer to these convenience premia as safety premia.

We first compare our estimates with the average convenience premium for French inflation-linked government bonds, known as OAT€s, reported in Christensen and Mouabbi (2024, henceforth CM). Their average estimated series is daily covering the period from October 2002 to December 2022 with a mean of 0.01 percent. Hence, there is a notable and persistent level difference in the real yields observed across the French and German inflation-linked bond markets thanks to the difference in their perceived safety. This result also means that we advise against pooling prices from these two markets, a practice used by economists at the European Central Bank (ECB) in the early years of these markets when the number of outstanding inflation-linked bonds in each was limited; see Ejlsing et al. (2007) for an example.³

We then compare our estimates with the safety premia of standard German bunds estimated by Christensen, Mirkov, and Zhang (2025, henceforth CMZ).⁴ Their estimated series average 1.24 percent over the 1999-2021 period. Hence, as expected, the estimated safety premia for the inflation-linked market are smaller, only about a quarter the size of those in the standard bund market.

Our findings also contrast with the results of ACR, who report an average estimated liquidity discount premium for U.S. Treasury Inflation-Protected Securities (TIPS) yields of 34 basis points for the 1997-2013 period. We speculate that the scarcity of both French and German inflation-linked bonds explains their high prices, whereas the U.S. TIPS market is much larger, better established, and therefore more likely to face the inherent liquidity challenges of inflation-linked bonds discussed at length in Cardozo and Christensen (2026).

Similarly to CM, we employ regression analysis to examine the determinants of the safety premia in the German inflation-linked government bond market. Using a large battery of explanatory variables, the results suggest that these premia behave less like liquidity premia and more like safety premia, which is in line with the regression results reported by CM for the convenience premia of French OAT€s.

We then dedicate a section to a detailed analysis of the market reaction to the announcement by the German Federal Finance Agency to cease all future issuance of inflation-linked debt made public on November 22, 2023. We find that neither the safety premia nor the trading conditions of the inflation-linked bonds were negatively affected by this decision. Thus, it does not appear that investors positioned themselves for somewhat slower inflation-linked market trading in the future. Rather, our analysis suggests that investors took it to signal

³We note that these differences may have mattered little for their sample because yield spreads across a wide variety of markets were very compressed in the years ahead of the Global Financial Crisis.

⁴CMZ provide estimates of safety premia for an international panel of government bond prices, including those of German bunds.

that the German government expected inflation to remain elevated causing inflation-linked debt not to be cost-effective. Overall, though, the market reaction was tempered and the inflation-linked market has continued to function on par with the past through the end of our sample. Hence, for now, German inflation-linked trading remains active despite no new issuance has come to market since before November 2023. These findings support our usage of the inflation-linked data through the end of our sample. They presumably will also support their usage well into the future, given that the longest-dated outstanding German inflation-linked bond can be expected to continue to trade until 2046. Although, at this point, it remains a rich source of information for both policy and trading analysis as we demonstrate in this paper, we caution that the usefulness of this market information will inevitably decline over time as the remaining inflation-linked bonds reach maturity.

The analysis in this paper relates to several important literatures. Most directly, our results relate to research on financial market liquidity and convenience premia. Second, our estimates of the real yield curve that would prevail without trading frictions have implications for asset pricing analysis on the true slope of the real yield curve. Furthermore, the paper is among the first to document what happens to trading and market dynamics when a government decides to terminate issuance of inflation-linked debt. Finally, the paper contributes to the rapidly growing literature on the economic consequences of the COVID-19 pandemic and its aftermath.

The remainder of the paper is organized as follows. Section 2 contains a description of the German inflation-linked government bond data, while Section 3 details the no-arbitrage term structure models we use and presents our empirical results. Section 4 describes the estimated German inflation-linked safety premia, including an analysis of their empirical determinants. Section 5 examines the German inflation-linked bond market response to the surprise cancellation of all future inflation-linked bond issuance. Finally, Section 6 concludes. Appendices contain details of additional bond data sets used in the analysis.

2 The German Inflation-Linked Government Bond Data

This section briefly describes the available data downloaded from Bloomberg for the market for German inflation-linked bonds referencing the harmonized index for consumer prices (HICP), officially known as Bund/€i's.⁵ Here, we will refer to them throughout as GBi's, and their nominal fixed-coupon equivalents as simply bunds, consistent with the literature.

To give a sense of the size of the German government bond market, we note up front that,

⁵We stress that these bonds are indexed using the euro area HICP index (ex. tobacco) without any seasonal adjustment, which is the benchmark for government bonds indexed to euro area inflation and the standard reference index for other financial products, most notably euro area inflation-linked swaps; see Ejsing et al. (2007).

Inflation-linked bond	Number of obs.		Issuance		Total uplifted notional amount
	Daily	Monthly	Date	amount	
(1) 1.50% 4/15/2016	1,943	90	3/15/2006	5,500	15,000
(2) 2.25% 4/15/2013	1,162	54	10/26/2007	4,000	11,000
(3) 1.75% 4/15/2020	2,558	118	6/12/2009	3,000	16,000
(4) 0.75% 4/15/2018	1,560	72	4/15/2011	3,000	15,000
(5) 0.10% 4/15/2023	2,613	121	3/23/2012	2,000	16,500
(6) 0.50% 4/15/2030	2,782	129	4/10/2014	2,000	22,150
(7) 0.10% 4/15/2026	2,542	118	3/12/2015	2,000	19,200
(8) 0.10% 4/15/2046	2,474	115	6/16/2015	500	14,250
(9) 0.10% 4/15/2033	1,006	47	2/11/2021	1,500	10,650

Table 1: **Sample of German Inflation-Linked Government Bonds**

The table reports the characteristics, first issuance date and amount, and total uplifted notional amount outstanding either at maturity or as of September 30, 2024, in millions of euros for the sample of German inflation-linked government bonds. Also reported are the number of daily and monthly observations for each bond during the sample period from October 26, 2007, to December 30, 2024.

as of the end of December 2023, the total outstanding notional amount of tradable securities issued by the German Federal government was €1.882 trillion, of which €66,25 billion, or 3.5 percent, represented inflation-linked bonds.⁶ Given the small size of the German government bond market relative to the German economy—with nominal GDP of €4.186 trillion in 2023—it is not surprising that the German government holds a triple-A rating with a stable outlook from all major rating agencies. Thus, there is effectively no credit risk to account for in the bond price data, and these bonds can be considered truly safe assets.

The German Federal government issued its first GBi bond on March 15, 2006, several years after France and Italy, which issued their first such government bonds in 2001 and 2003, respectively.⁷ Moreover, the German inflation-linked market is characterized by a very limited number of bonds, most of which have had ten years or less time to maturity at issuance. These relatively short maturities set the German market apart from other inflation-linked bond markets—with the exception of Japan, where the government solely issues ten-year inflation-linked bonds; see Christensen and Spiegel (2022).

Table 1 contains the contractual details of all nine GBi's, as well as the number of daily and monthly observations for each, while the time-varying maturity distribution of the nine GBi's in our sample is illustrated in Figure 1. Here, each security is represented by a downward-sloping line showing its remaining years to maturity at each date. The limited set of bonds

⁶This information is available at <https://www.deutsche-finanzagentur.de/en/federal-securities/trading/tradeable-securities> and <https://www.deutsche-finanzagentur.de/en/federal-securities/types-of-federal-securities/inflation-linked-federal-securities>.

⁷All auction information for German federal government securities back to January 1999 is available at: <https://www.deutsche-finanzagentur.de/en/federal-securities/issuances/issuance-results>.

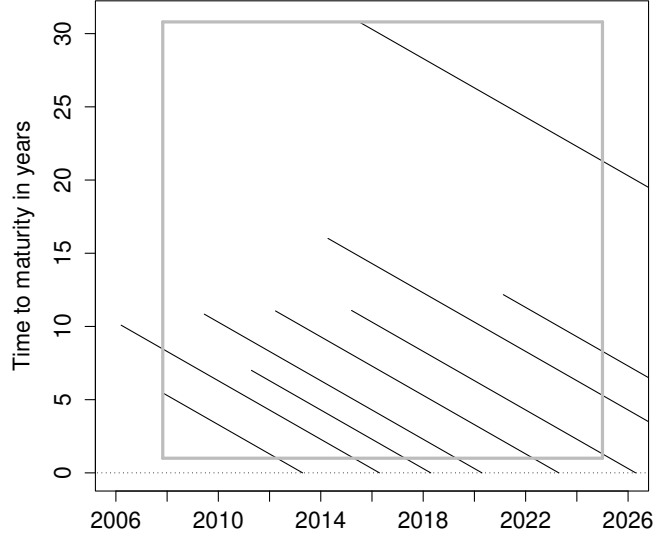


Figure 1: Maturity Distribution of German Inflation-Linked Government Bonds
 Illustration of the maturity distribution of the available universe of German inflation-linked government bonds. The solid grey rectangle indicates the sample used in the empirical analysis, where the sample is restricted to start on October 26, 2007, and end on December 30, 2024, and limited to bond prices with more than one year to maturity.

poses some challenges in modeling the term structure of interest rates in this market, but we use recently developed tools to deal with this technical complication. Moreover, these features combined also explain why few papers have examined this market in detail.

Figure 2 shows the yields to maturity for all nine German GBi bonds in our sample at daily frequency from October 26, 2007, to December 30, 2024.⁸ Note the following regarding these yield series. First, we highlight the significant persistent decline in real yields over the first fifteen years of the sample, as well as the notable sharp partial reversal during the last three years of the sample. Long-term real yields in the euro area were close to 2 percent in late 2007 and had dropped below -2 percent by late 2021 before retracing more than half of that decline by the end of our sample. Second, business cycle variations in the shape of the yield curve are pronounced around the persistent trends. The yield curve tends to flatten ahead of recessions and steepen during the initial phase of economic recoveries. These characteristics are the practical motivation behind our choice of using a three-factor model for the frictionless part of the German real yield curve, adopting an approach similar to what is standard for

⁸Our model estimation requires at least two observed bond prices for each observation date. This dictates the start date for our sample.

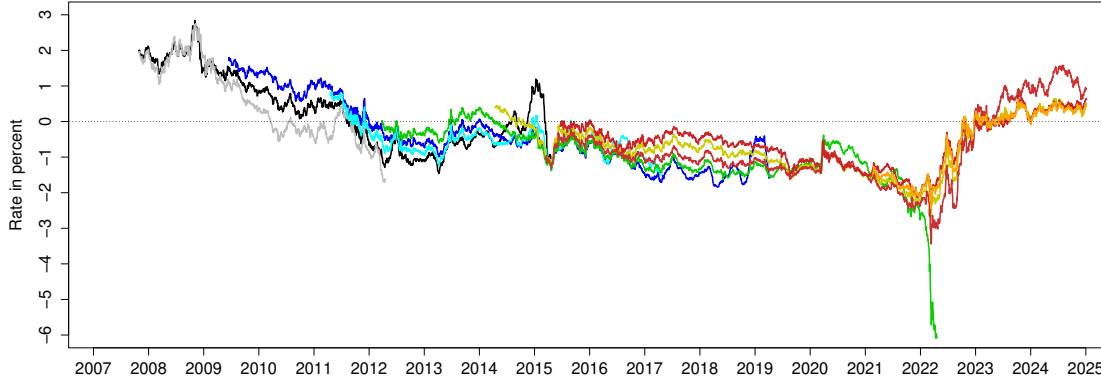


Figure 2: Yield to Maturity of German Inflation-Linked Government Bonds

Illustration of the yield to maturity implied by the German inflation-linked government bond prices considered in this paper, which are subject to two sample choices: (1) sample limited to the period from October 26, 2007, to December 30, 2024; (2) censoring of a bond's price when it has less than one year to maturity. Each bond yield series is shown with its own colored line.

Bond maturity	Banks	Government	Households	Insurance	NFCs	Non-MMF	Pensions	Other
Below 10 years	10.02	14.19	6.72	12.02	2.58	44.71	7.31	2.45
11 to 20 years	5.94	13.59	4.19	13.21	1.56	39.66	21.42	0.45
Above 20 years	2.14	2.14	0.75	9.41	0.27	26.27	52.13	0.48

Table 2: Sectoral Holdings of German GBi Bonds

The table reports the average share of holdings (in percent) of German GBi bonds, across sectors, by tenor segment. These statistics are produced using Eurosystem SHSS data for the period 2013Q4-2024Q4. NFCs are non-financial corporations and Non-MMFs are non-money market investment funds.

U.K. and U.S. nominal yield data; see Christensen and Rudebusch (2012).

To better understand the demand for German GBi bonds, we exploit euro-area security holdings data at the sector level, which covers the entire universe of holders of euro-area securities. On average for the period from 2013Q4 to 2024Q4, non-money market investment funds (Non-MMFs), pension funds, governments, and insurance companies were the largest holders of German GBi bonds, with respective average holdings of about 38 percent, 25 percent, 13 percent, and 12 percent. Table 2 reports the average share of holdings by sectors, across three maturity segments. For GBi bonds with maturities up to 20 years, non-money market investment funds (Non-MMFs) hold the largest share. In contrast, for the single GBi with maturity above 20 years, pension funds had the largest average ownership share slightly above 50 percent, followed by Non-MMFs, insurance companies, and banks. Moreover, on average, about 26 percent of these bonds were held by German investors, with the rest being

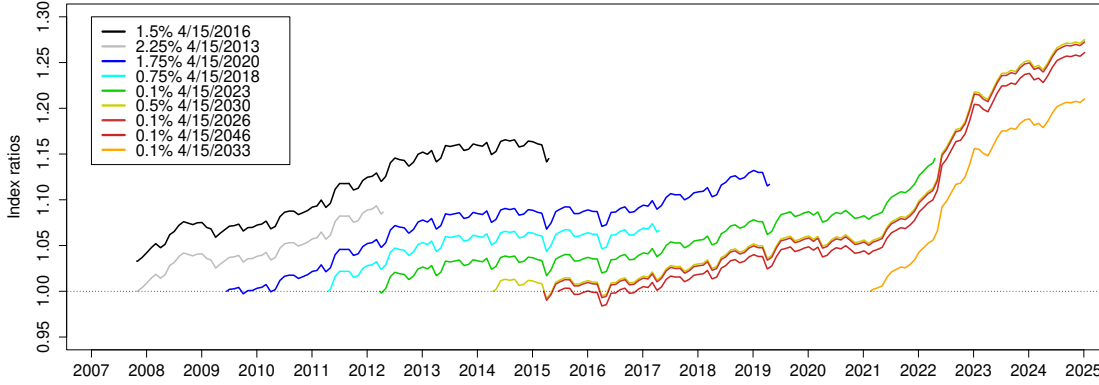


Figure 3: Index Ratios of German Inflation-Linked Government Bonds

Illustration of the index ratios of the German inflation-linked government bonds considered in this paper, which are subject to two sample choices: (1) sample limited to the period from October 26, 2007, to December 30, 2024; (2) censoring of a bond's index ratio when it has less than one year to maturity.

held abroad. Prominent investors outside of Germany included the Netherlands (31 percent), France (13 percent), Luxembourg (10 percent) and Ireland (9 percent).

Figure 3 shows the inflation index ratios for all 9 GBi bonds in our sample. We note that none of the bonds have been exposed to any prolonged period of deflation, defined as periods with inflation index ratios below one. Indeed, thanks to the generally positive inflation environment in the euro area, the ratios tend to relatively quickly become significantly positive. This suggests that their offered deflation protection is likely to be of modest value, similar to what Christensen and Mouabbi (2023) find for French government bonds indexed using the French CPI and known as OATi's. We therefore disregard this component in our analysis and leave it for future research to assess its value. Also notable is the seasonality in the HICP (ex tobacco). However, thanks to the relatively long maturities of the bonds in our sample, the price impact of the seasonality will tend to be relatively modest; see D'Amico et al. (2018).⁹

Before turning to our models and their estimation, we examine the bid-ask spreads of the German inflation-linked government bonds to provide support for the ACR approach to identifying the bond-specific risk premia. The spreads are constructed by converting the bid and ask prices into the corresponding yield to maturity and calculating the difference with all data downloaded from Bloomberg. Figure 4 shows the bid-ask spread series for all nine GBi's since February 2011, when the data become available on Bloomberg. All series are smoothed four-week moving averages measured in basis points. Similar to what ACR document for U.S.

⁹See Ejlsing et al. (2007) for an analysis of the seasonality effects in the euro-area setting.

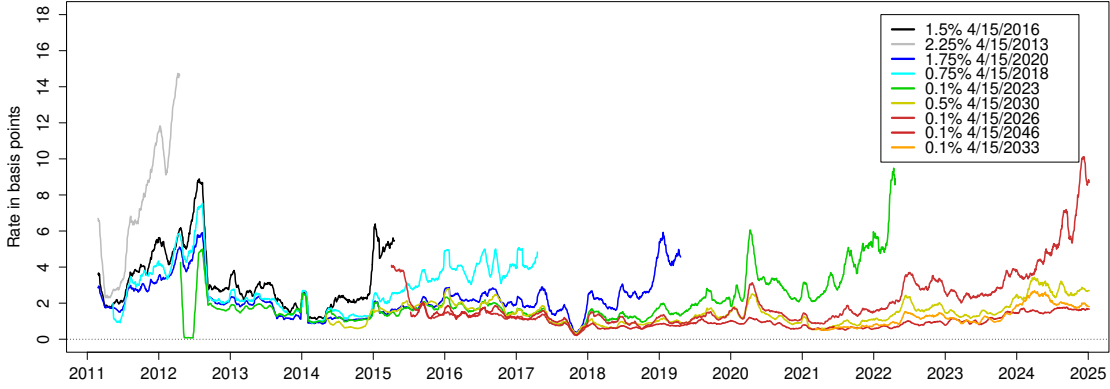


Figure 4: **Bid-Ask Spreads of German Inflation-Linked Government Bonds**

Illustration of the four-week moving average of bid-ask spreads of German inflation-linked government bonds constructed as explained in the main text. The series are daily covering the period from February 22, 2011, to December 30, 2024.

TIPS, the German GBi bid-ask spreads are systematically wider for more seasoned bonds than for recently issued bonds. Rational, forward-looking investors are aware of these dynamics and the fact that future market liquidity of a given bond is likely to be below its current market liquidity. This gives rise to bond-specific premia in the bond prices. This pattern in observed measures of *current* market liquidity of German GBi's is consistent with the factor loading of the bond-specific risk factor in our approach that is intended to model the effects on current GBi prices of expected *future* market demand conditions. Although the natural interpretation of these premia would be to think of them as liquidity discounts, we note that, given the high credit quality of these bonds, they may be viewed by investors as very safe assets and hence trade at a safety premium; see Christensen and Mirkov (2022). We stress that the model we use is flexible enough to accommodate either of these outcomes.

3 Model Estimation and Results

In this section, we first describe how we model yields in a world without any frictions to trading. This model of frictionless dynamics is fundamental to our analysis. We then detail the augmented model that accounts for the bond-specific premia in the inflation-linked bond yields. This is followed by a description of the restrictions imposed to achieve econometric identification of this model and its estimation. We end the section with a brief summary of our estimation results.

3.1 A Frictionless Arbitrage-Free Model of Real Yields

To capture the fundamental or frictionless factors operating the German GBi real yield curve, we choose to focus on the tractable affine dynamic term structure model introduced in Christensen et al. (2011).¹⁰

In this arbitrage-free Nelson-Siegel (AFNS) model, the state vector is denoted by $X_t = (L_t, S_t, C_t)$, where L_t is a level factor, S_t is a slope factor, and C_t is a curvature factor. The instantaneous risk-free real rate is defined as

$$r_t = L_t + S_t. \quad (1)$$

The risk-neutral (or $\mathbb{Q}$) dynamics of the state variables used for pricing are given by the stochastic differential equations¹¹

$$\begin{pmatrix} dL_t \\ dS_t \\ dC_t \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -\lambda & \lambda \\ 0 & 0 & -\lambda \end{pmatrix} \begin{pmatrix} L_t \\ S_t \\ C_t \end{pmatrix} dt + \Sigma \begin{pmatrix} dW_t^{L,\mathbb{Q}} \\ dW_t^{S,\mathbb{Q}} \\ dW_t^{C,\mathbb{Q}} \end{pmatrix}, \quad (2)$$

where Σ is the constant covariance (or volatility) matrix that is assumed to be diagonal, as recommended by Christensen et al. (2011).¹² Based on this specification of the $\mathbb{Q}$ -dynamics, real zero-coupon bond yields preserve the Nelson-Siegel factor loading structure as

$$y_t(\tau) = L_t + \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} \right) S_t + \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right) C_t - \frac{A(\tau)}{\tau}, \quad (3)$$

where $A(\tau)$ is a convexity term that adjusts the functional form in Nelson and Siegel (1987) to ensure absence of arbitrage; see Christensen et al. (2011).

To complete the description of the model and to implement it empirically, we will need to specify the risk premia that connect these factor dynamics under the $\mathbb{Q}$ -measure to the dynamics under the real-world (or physical) $\mathbb{P}$ -measure. It is important to note that there are no restrictions on the dynamic drift components under the empirical $\mathbb{P}$ -measure beyond the requirement of constant volatility. To facilitate empirical implementation, we use the essentially affine risk premium specification introduced in Duffee (2002). In the Gaussian framework, this specification implies that the risk premia Γ_t depend on the state variables;

¹⁰Although the model is not formulated using the canonical form of affine term structure models introduced by Dai and Singleton (2000), it can be viewed as a restricted version of the canonical Gaussian model; see Christensen et al. (2011) for details.

¹¹As discussed in Christensen et al. (2011), with a unit root in the level factor, the model is not arbitrage-free with an unbounded horizon; therefore, as is often done in theoretical discussions, we impose an arbitrary maximum horizon.

¹²As per Christensen et al. (2011), $\theta^{\mathbb{Q}}$ is set to zero without loss of generality.

that is,

$$\Gamma_t = \gamma^0 + \gamma^1 X_t,$$

where $\gamma^0 \in \mathbf{R}^3$ and $\gamma^1 \in \mathbf{R}^{3 \times 3}$ contain unrestricted parameters.

Thus, the resulting unrestricted three-factor AFNS model has $\mathbb{P}$ -dynamics given by

$$\begin{pmatrix} dL_t \\ dS_t \\ dC_t \end{pmatrix} = \begin{pmatrix} \kappa_{11}^{\mathbb{P}} & \kappa_{12}^{\mathbb{P}} & \kappa_{13}^{\mathbb{P}} \\ \kappa_{21}^{\mathbb{P}} & \kappa_{22}^{\mathbb{P}} & \kappa_{23}^{\mathbb{P}} \\ \kappa_{31}^{\mathbb{P}} & \kappa_{32}^{\mathbb{P}} & \kappa_{33}^{\mathbb{P}} \end{pmatrix} \left(\begin{pmatrix} \theta_1^{\mathbb{P}} \\ \theta_2^{\mathbb{P}} \\ \theta_3^{\mathbb{P}} \end{pmatrix} - \begin{pmatrix} L_t \\ S_t \\ C_t \end{pmatrix} \right) dt + \Sigma \begin{pmatrix} dW_t^{L,\mathbb{P}} \\ dW_t^{S,\mathbb{P}} \\ dW_t^{C,\mathbb{P}} \end{pmatrix}.$$

This is the transition equation in the Kalman filter estimation.

3.2 An Arbitrage-Free Model of Real Yields with Bond-Specific Risk

In this section, we augment the frictionless AFNS model introduced above to account for any bond-specific risk premia embedded in the GBi prices. To do so, let $X_t = (L_t, S_t, C_t, X_t^R)$ denote the state vector of the four-factor AFNS-R model with bond-specific risk premium adjustment. As in the non-augmented model, we let the frictionless instantaneous real risk-free rate be defined by equation (1), while the risk-neutral dynamics of the state variables used for pricing are given by

$$\begin{pmatrix} dL_t \\ dS_t \\ dC_t \\ dX_t^R \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \lambda & -\lambda & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & \kappa_R^{\mathbb{Q}} \end{pmatrix} \left[\begin{pmatrix} 0 \\ 0 \\ 0 \\ \theta_R^{\mathbb{Q}} \end{pmatrix} - \begin{pmatrix} L_t \\ S_t \\ C_t \\ X_t^R \end{pmatrix} \right] dt + \Sigma \begin{pmatrix} dW_t^{L,\mathbb{Q}} \\ dW_t^{S,\mathbb{Q}} \\ dW_t^{C,\mathbb{Q}} \\ dW_t^{R,\mathbb{Q}} \end{pmatrix},$$

where Σ continues to be a diagonal matrix.

In the augmented model, GBi yields are sensitive to bond-specific risks because the net present value of their future cash flow is calculated using the following discount function:

$$\bar{r}^i(t, t_0^i) = r_t + \beta^i(1 - e^{-\lambda^{R,i}(t-t_0^i)})X_t^R = L_t + S_t + \beta^i(1 - e^{-\lambda^{R,i}(t-t_0^i)})X_t^R. \quad (4)$$

Christensen and Rudebusch (2019) show that the net present value of one unit of consumption paid by GBi bond i at time $t + \tau$ has the following exponential-affine form

$$\begin{aligned} P_t(t_0^i, \tau) &= E^{\mathbb{Q}} \left[e^{-\int_t^{t+\tau} \bar{r}^i(s, t_0^i) ds} \right] \\ &= \exp \left(B_1(\tau)L_t + B_2(\tau)S_t + B_3(\tau)C_t + B_4(t, t_0^i, \tau)X_t^R + A(t, t_0^i, \tau) \right). \end{aligned}$$

This result implies that the model belongs to the class of Gaussian affine term structure models. Note also that, by fixing $\beta^i = 0$ for all i , we recover the AFNS model.

Now, consider the whole value of GBi bond i issued at time t_0^i with maturity at $t + \tau^i$ that pays an annual coupon C^i . Its price is given by¹³

$$\begin{aligned} \bar{P}_t(t_0^i, \tau^i, C^i) &= C^i(t_1 - t)E^{\mathbb{Q}}\left[e^{-\int_t^{t_1} \bar{r}^{R,i}(s, t_0^i)ds}\right] + \sum_{j=2}^N C^i E^{\mathbb{Q}}\left[e^{-\int_t^{t_j} \bar{r}^{R,i}(s, t_0^i)ds}\right] \\ &\quad + E^{\mathbb{Q}}\left[e^{-\int_t^{t+\tau^i} \bar{r}^{R,i}(s, t_0^i)ds}\right]. \end{aligned}$$

There are only two minor omissions in this bond pricing formula. First, it does not account for the lag in the inflation indexation of the GBi bond payoff. The potential error from this omission should be modest (see Grishchenko and Huang 2013), especially as we exclude bonds from our sample when they have less than one year of maturity remaining. Second, we do not account for the value of deflation protection offered by GBi bonds, as already noted. However, Christensen and Mouabbi (2023) find these values to be very small for French OATi indexed to the French consumer price index, and, given that HICP inflation has run quite a bit above French CPI inflation during our sample, the value of this protection for GBi bonds is likely to be tiny as well.

Finally, to complete the description of the AFNS-R model, we again specify an essentially affine risk premium structure, which implies that the risk premia Γ_t take the form

$$\Gamma_t = \gamma^0 + \gamma^1 X_t,$$

where $\gamma^0 \in \mathbf{R}^4$ and $\gamma^1 \in \mathbf{R}^{4 \times 4}$ contain unrestricted parameters. Thus, the resulting unrestricted four-factor AFNS-R model has $\mathbb{P}$ -dynamics given by

$$\begin{pmatrix} dL_t \\ dS_t \\ dC_t \\ dX_t^R \end{pmatrix} = \begin{pmatrix} \kappa_{11}^{\mathbb{P}} & \kappa_{12}^{\mathbb{P}} & \kappa_{13}^{\mathbb{P}} & \kappa_{14}^{\mathbb{P}} \\ \kappa_{21}^{\mathbb{P}} & \kappa_{22}^{\mathbb{P}} & \kappa_{23}^{\mathbb{P}} & \kappa_{24}^{\mathbb{P}} \\ \kappa_{31}^{\mathbb{P}} & \kappa_{32}^{\mathbb{P}} & \kappa_{33}^{\mathbb{P}} & \kappa_{34}^{\mathbb{P}} \\ \kappa_{41}^{\mathbb{P}} & \kappa_{42}^{\mathbb{P}} & \kappa_{43}^{\mathbb{P}} & \kappa_{44}^{\mathbb{P}} \end{pmatrix} \left(\begin{pmatrix} \theta_1^{\mathbb{P}} \\ \theta_2^{\mathbb{P}} \\ \theta_3^{\mathbb{P}} \\ \theta_4^{\mathbb{P}} \end{pmatrix} - \begin{pmatrix} L_t \\ S_t \\ C_t \\ X_t^R \end{pmatrix} \right) dt + \Sigma \begin{pmatrix} dW_t^{L, \mathbb{P}} \\ dW_t^{S, \mathbb{P}} \\ dW_t^{C, \mathbb{P}} \\ dW_t^{R, \mathbb{P}} \end{pmatrix}.$$

This is the transition equation in the Kalman filter estimation.

3.3 Model Estimation and Econometric Identification

Due to the nonlinear relationship between the state variables and the bond prices, the model cannot be estimated with the standard Kalman filter. Instead, we use the extended Kalman filter as in Kim and Singleton (2012); see Christensen and Rudebusch (2019) for details. Furthermore, to make the fitted errors comparable across bonds of various maturities, we

¹³This is the clean price that does not account for any accrued interest and maps to our observed bond prices.

scale each bond price by its duration. Thus, the measurement equation for the bond prices takes the following form

$$\frac{P_t^i(t_0^i, \tau^i)}{D_t^i(t_0^i, \tau^i)} = \frac{\hat{P}_t^i(t_0^i, \tau^i)}{D_t^i(t_0^i, \tau^i)} + \varepsilon_t^i,$$

where $\hat{P}_t^i(t_0^i, \tau^i)$ is the model-implied price of bond i and $D_t^i(t_0^i, \tau^i)$ is its duration, which is calculated before estimation. See Andreasen et al. (2019) for evidence supporting this formulation of the measurement equation.

Furthermore, to ensure econometric identification of the β^i loadings on the bond-specific risk factor, we let the first GBi issued on March 15, 2006, have a unit loading on this factor, that is, this ten-year bond maturing on April 15, 2016, with 1.50 percent coupon has $\beta^i = 1$. This choice implies that the β^i sensitivity parameters measure bond-specific risk sensitivity relative to that of the ten-year 2016 GBi.

Finally, we note that the $\lambda^{R,i}$ parameters can be hard to identify if their values are too large or too small. As a consequence, we follow ACR and impose the restriction that they fall within the range from 0.0001 to 10, which is without practical consequences.¹⁴ Also, for numerical stability during model optimization, we impose the restriction that the β^i parameters fall within the range from 0 to 250, which turns out not to be a binding constraint for any of the nine bonds in our sample. Hence, this constraint is also without practical consequences.

As is standard in the finance literature, we assume that the state variables are stationary, which allows us to start the Kalman filter at their unconditional mean and covariance matrix.

3.4 Estimation Results

This section presents our benchmark estimation results. In the interest of simplicity, we focus on a version of the AFNS-R model where $K^{\mathbb{P}}$ and Σ are diagonal matrices. As shown in ACR, these restrictions have only small effects on the estimated bond-specific risk premium for each inflation-linked bond, because it is identified from the model's $\mathbb{Q}$ -dynamics, which are independent of $K^{\mathbb{P}}$ and only display a weak link to Σ through the small convexity adjustment in the bond yields.

Table 3 reports the summary statistics for the fitted errors of individual GBi bonds as well as for all bonds combined. Note that there is uniform improvement in model fit from incorporating the bond-specific risk factor into the AFNS model. Still, it is worth noting that the AFNS model is able to deliver a root mean-squared fitted error of 3.9 basis points across all bonds combined, which in general could be characterized as a satisfactory fit, but obviously not as good as the RMSE of 3.0 basis points for all bonds combined achieved by

¹⁴To see this, note that the involved term, $e^{-\lambda^{R,i}(t-t_0^i)}$, becomes really tiny for values of $\lambda^{R,i}$ above 10. As a consequence, allowing $\lambda^{R,i}$ to be even larger would matter little for our reported results; see Christensen et al. (2021) for evidence.

GBi bond	Pricing errors				Estimated parameters			
	AFNS		AFNS-R		AFNS-R			
	Mean	RMSE	Mean	RMSE	β^i	SE	$\lambda^{R,i}$	SE
(1) 1.50% 4/15/2016	0.46	3.26	0.36	2.08	1	n.a.	0.1736	0.0377
(2) 2.25% 4/15/2013	-0.40	2.95	-0.05	2.17	0.9213	0.1343	3.1341	3.3628
(3) 1.75% 4/15/2020	0.27	2.74	-0.02	1.33	80.0536	6.2566	0.0008	0.0000
(4) 0.75% 4/15/2018	0.25	3.16	0.02	1.73	0.5622	0.0522	8.6554	5.6077
(5) 0.10% 4/15/2023	-0.32	7.12	-0.08	6.80	42.6442	4.8381	0.0018	0.0001
(6) 0.50% 4/15/2030	-0.54	2.81	0.73	1.53	0.6764	0.2152	0.5050	0.2342
(7) 0.10% 4/15/2026	1.22	3.70	0.92	1.65	0.9447	0.1915	0.1973	0.0617
(8) 0.10% 4/15/2046	1.32	3.16	0.85	1.46	0.1745	0.4255	10.0000	6.8646
(9) 0.10% 4/15/2033	1.77	3.17	0.78	1.92	0.5354	0.2605	10.0000	6.0020
All yields	0.39	3.94	0.41	2.99	-	-	-	-
Max $\mathcal{L}^{EKF}$	4,434.62		4,698.03		-		-	

Table 3: **Pricing Errors and Estimated Bond-Specific Risk Parameters**

This table reports the mean pricing errors (Mean) and the root mean-squared pricing errors (RMSE) of German inflation-linked bonds in the AFNS and AFNS-R models estimated with a diagonal specification of $K^{\mathbb{P}}$ and Σ . The errors are computed as the difference between the German GBi bond market price expressed as yield to maturity and the corresponding model-implied yield. All errors are reported in basis points. Standard errors (SE) are not available (n.a.) for the normalized value of β^1 . The data are monthly covering the period from October 31, 2007, to December 30, 2024.

the AFNS-R model, which represents a really good fit to the entire cross section of yields. This salient fit is also on display in Figure 5, which shows the individual fitted error series from the AFNS-R model. With the exception of a single observation from a single bond, the model consistently provides a very accurate fit to the entire cross section of bond prices.¹⁵

Table 4 contains the estimated dynamic parameters. Note that the dynamics of the first three factors are rather different across the two estimations. Although the estimated mean parameters are comparable for the first three factors, their estimated mean-reversion and volatility parameters are notably larger for the slope and curvature factor in the AFNS model. Hence, the frictionless dynamics of the state variables within the AFNS-R model are broadly somewhat more persistent and less volatile. Furthermore, λ is smaller in the AFNS-R model. This implies that the yield loadings of the slope factor decays toward zero more slowly as the maturity increases. At the same time, the peak of the curvature yield loadings is located at a later maturity compared with its loading in the AFNS model. As a consequence, slope and curvature matter more for longer-term yields in the AFNS-R model. This helps explain part of the better fit to the entire cross section of bonds within that model. Finally, the bond-specific risk factor is the least persistent and most volatile factor in the AFNS-R

¹⁵The large single error is for the 0.1% 4/15/2023 GBi and coincides with Russia’s invasion of Ukraine in late February 2022. Because the bond is close to maturity, it is very sensitive to short-term inflationary market shocks of that kind, which are poorly captured within our model as its emphasis is on fitting medium- and long-term yields.

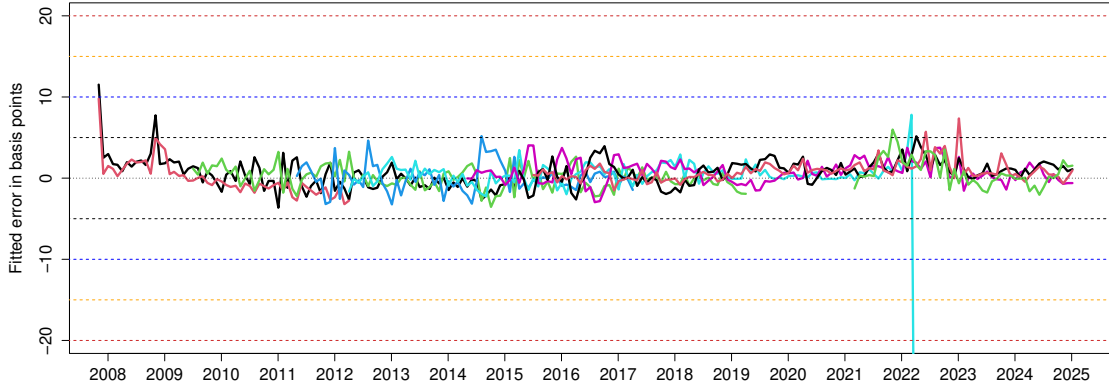


Figure 5: **Fitted Errors of GBi Bond Yields**

Illustration of the fitted errors of GBi yields to maturity implied by the AFNS-R model with a diagonal specification of $K^{\mathbb{P}}$ and Σ estimated at monthly frequency for the period from October 31, 2007, to December 30, 2024.

model. Moreover, it is even more stationary under the risk-neutral $\mathbb{Q}$ -measure used for pricing and with a negative mean equal to -0.0060. The high value of $\kappa_R^{\mathbb{Q}}$ combined with the negative value of $\theta_R^{\mathbb{Q}}$ suggests that the bond-specific risk premia are likely to be mostly negative, a conjecture we confirm in the following section.

The estimated paths of the level, slope, and curvature factors from the two models are shown in Figure 6. While the two models' level factors are fairly close to each other most of the time, their slope and curvature factors tend to have wedges between them. However, they generally operate at similar levels and frequently move in tandem. Hence, the roles of these three factors within each model can be characterized as broadly similar. Importantly, there are some sharp spikes in the data that are ascribed to the slope and curvature factors within the AFNS model, while they appear to be captured by the bond-specific risk factor within the AFNS-R model. This also helps explain why the slope and curvature factors are less volatile within the AFNS-R model compared to the AFNS model.

4 The GBi Bond-Specific Risk Premium

In this section, we analyze the German GBi bond-specific risk premia implied by the estimated AFNS-R model described in the previous section. First, we formally define the bond-specific risk premium, study its historical evolution, and compare it with other estimates from the literature before we briefly assess its sensitivity to the assumed factor dynamics and the data frequency used in the model estimation. We then perform one regression analysis that

Parameter	AFNS		AFNS-R	
	Est.	SE	Est.	SE
$\kappa_{11}^{\mathbb{P}}$	0.0433	0.0243	0.2044	0.1692
$\kappa_{22}^{\mathbb{P}}$	0.8603	0.0273	0.3857	0.3097
$\kappa_{33}^{\mathbb{P}}$	1.4287	0.0343	0.4492	0.2857
$\kappa_{44}^{\mathbb{P}}$	-	-	1.1784	0.5314
σ_{11}	0.0045	0.0001	0.0072	0.0004
σ_{22}	0.0215	0.0014	0.0132	0.0014
σ_{33}	0.0274	0.0020	0.0143	0.0018
σ_{44}	-	-	0.2445	0.1652
$\theta_1^{\mathbb{P}}$	0.0224	0.0131	0.0181	0.0087
$\theta_2^{\mathbb{P}}$	-0.0083	0.0072	-0.0074	0.0094
$\theta_3^{\mathbb{P}}$	-0.0341	0.0063	-0.0226	0.0097
$\theta_4^{\mathbb{P}}$	-	-	0.0280	0.0666
λ	0.3987	0.0053	0.3055	0.0294
$\kappa_R^{\mathbb{Q}}$	-	-	9.7892	6.5212
$\theta_R^{\mathbb{Q}}$	-	-	-0.0060	0.0015
σ_y	0.0005	1.69×10^{-5}	0.0002	1.75×10^{-5}

Table 4: **Estimated Dynamic Parameters**

The table shows the estimated dynamic parameters for the AFNS and AFNS-R models estimated with a diagonal specification of $K^{\mathbb{P}}$ and Σ . The data are monthly covering the period from October 31, 2007, to December 30, 2024.

examines the determinants of the GBi bond-specific risk premia in general and another that focuses narrowly on their response to the ECB’s large-scale bond purchases during the 2015-2021 period. We end the section detailing how measures of breakeven inflation are affected by adjustments for the large GBi safety premia.

4.1 The Estimated GBi Bond-Specific Risk Premia

We now use the estimated AFNS-R model to extract the bond-specific risk premium in the GBi market. To compute these premia, we first use the estimated parameters and the filtered states $\{X_{t|t}\}_{t=1}^T$ to calculate the fitted GBi prices $\{\hat{P}_t^i\}_{t=1}^T$ for all outstanding GBi securities in our sample. These bond prices are then converted into yields to maturity $\{\hat{y}_t^{c,i}\}_{t=1}^T$ by solving the fixed-point problem

$$\begin{aligned}
\hat{P}_t^i = & C(t_1 - t) \exp \left\{ -(t_1 - t) \hat{y}_t^{c,i} \right\} + \sum_{k=2}^n C \exp \left\{ -(t_k - t) \hat{y}_t^{c,i} \right\} \\
& + \exp \left\{ -(T - t) \hat{y}_t^{c,i} \right\},
\end{aligned} \tag{5}$$

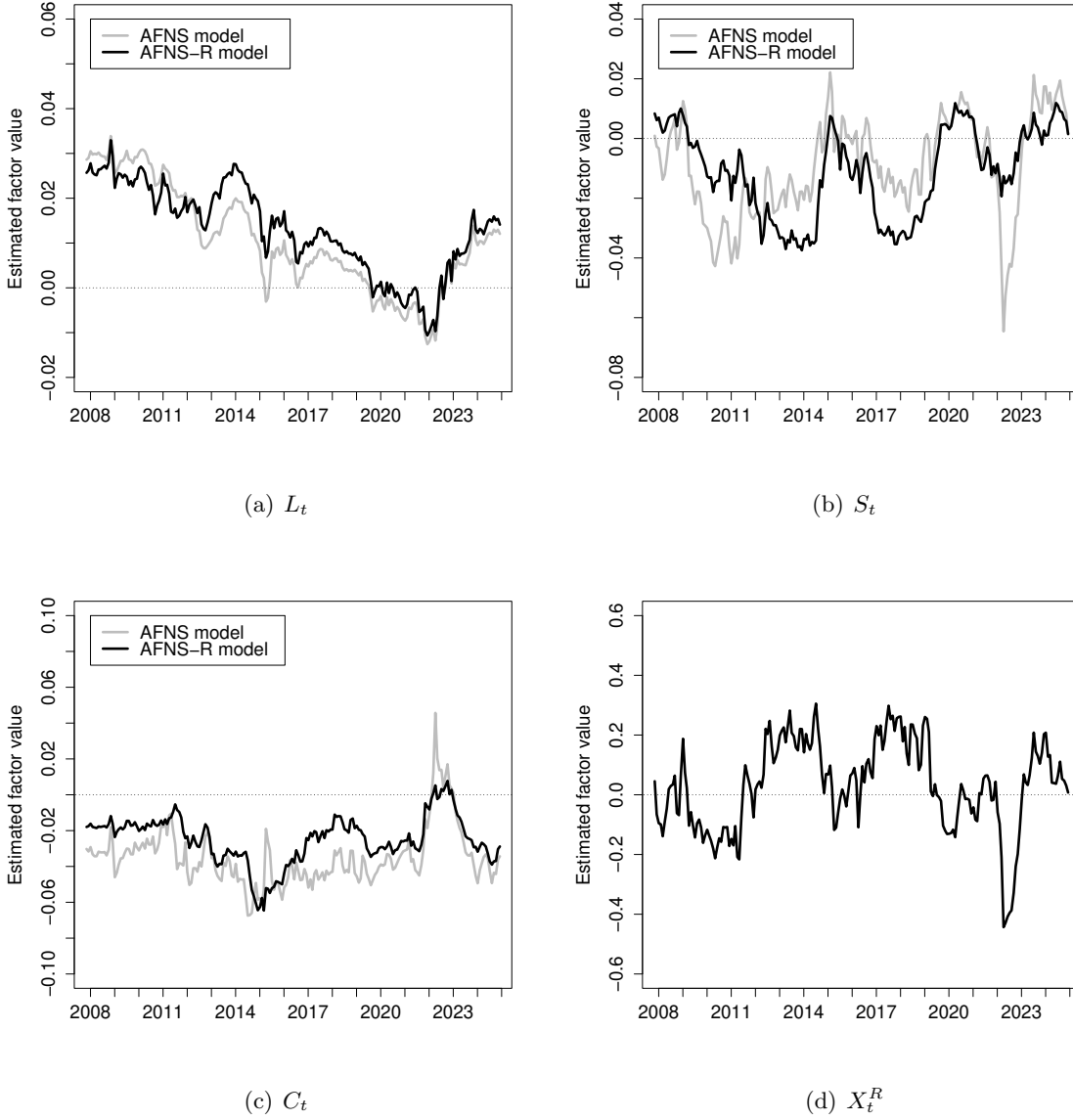


Figure 6: Estimated State Variables

Illustration of the estimated state variables from the AFNS and AFNS-R models, each with a diagonal specification of $K^{\mathbb{P}}$ and Σ and estimated at monthly frequency for the period from October 31, 2007, to December 30, 2024.

for $i = 1, 2, \dots, n_{GBi}$, meaning that $\left\{ \hat{y}_t^{c,i} \right\}_{t=1}^T$ is approximately the real rate of return on the i th GBi bond if held until maturity (see Sack and Elsassser 2004). To obtain the corresponding yields with correction for the bond-specific risk premia, we compute a new set of model-implied bond prices from the estimated AFNS-R model using only its frictionless part, i.e., using the constraints that $X_{t|t}^R = 0$ for all t as well as $\sigma_{44} = 0$ and $\theta_R^Q = 0$. These prices are denoted $\left\{ \tilde{P}_t^i \right\}_{t=1}^T$ and converted into yields to maturity $\tilde{y}_t^{c,i}$ using equation (5). They

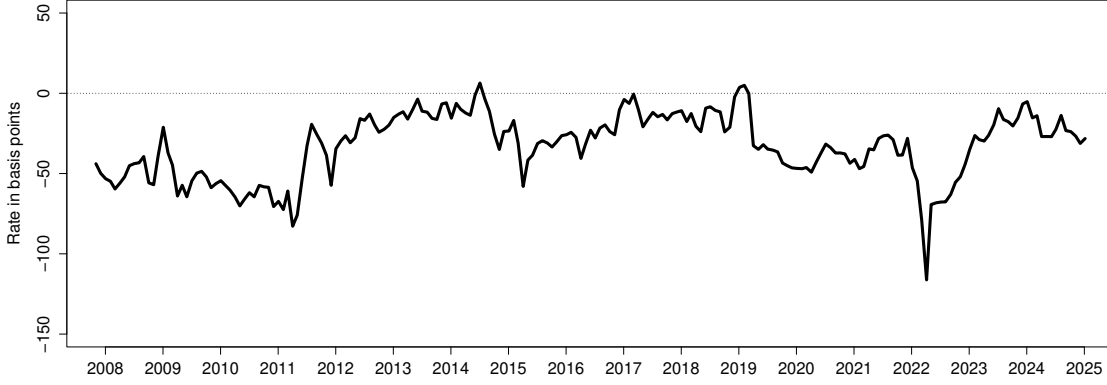


Figure 7: **Average Estimated GBI Bond-Specific Risk Premium**

Illustration of the average estimated bond-specific risk premium of GBI's for each observation date implied by the AFNS-R model estimated with a diagonal specification of $K^{\mathbb{P}}$ and Σ . The bond-specific risk premia are measured as the estimated yield difference between the fitted yield to maturity of individual GBI's and the corresponding frictionless yield to maturity with the bond-specific risk factor turned off. The data are monthly and cover the period from October 31, 2007, to December 30, 2024.

represent estimates of the prices that would prevail in a world without any financial frictions or special demands for certain bonds. The bond-specific risk premium for the i th GBI bond is then defined as

$$\Psi_t^i \equiv \hat{y}_t^{c,i} - \tilde{y}_t^{c,i}. \quad (6)$$

Figure 7 shows the average estimated GBI bond-specific risk premium $\bar{\Psi}_t$ across the outstanding GBI bonds at each point in time. Note that a negative value means that the fitted GBI bond price is *above* the model-implied frictionless price, i.e., GBI bond prices are higher than they should be in a world without any frictions. Importantly, the mean of the series is -0.33 percent. Thus, on average, GBI bond prices are higher than they are likely to be in a frictionless world without any excess demand for safe assets. Given the relatively low liquidity of these bonds, this convenience premium cannot be a consequence of their moneyiness as it is challenging to trade these bonds in large volumes on short notice. Instead, we follow Christensen and Mirkov (2022) and interpret it as a safety premium investors are willing to pay thanks to the high credit quality of these bonds. Moreover, there are some detectable trends and time variation in the series, which explains its standard variation of 20.13 basis points. These safety premia were increasing and approaching zero on average during the European sovereign debt crisis 2011-2013. This suggests that even German government bonds were perceived as less-safe debt instruments during that challenging period. The average premium

then trended sideways slightly below zero until 2019. It then experienced a pronounced and persistent decline that lasted until spring 2022. Hence, according to our model, a notable part of the decline in GBI yields during the pandemic years reflected declines in the bond-specific safety premia. We take this to indicate that GBI's regained their status as a very safe class of bonds. Notably, the average safety premium got a boost and dropped below -100 basis points in March 2022 when HICP inflation was highly elevated. We interpret this drop as an added convenience premium arising from the fact that, when inflation is highly elevated, inflation-linked bonds like GBI's become convenient assets to hold. This boost was short-lived, though, as ECB and other central banks responded forcefully to the inflation spike by tightening monetary policy significantly. We take the relatively quick normalization of the estimated safety premium series as a sign that investors did not expect the high inflation to persist for very long, an indicator of central bank credibility of sorts. As a consequence, during the remaining part of our sample, the average estimated bond-specific safety premium was close to its historical average.

To summarize, the average estimated GBI bond-specific safety premium broadly follows a reasonable time series pattern that aligns well with the safety premium interpretation that we adopt henceforth.

As an additional validation exercise and to put our average estimated safety premium from the market for German GBI's into an international context, we compare it with similar estimates from two other major bond markets, specifically the market for French OAT€s with cash flows also adjusted to the HICP (ex. tobacco) examined by CM and the much larger market for standard German bunds studied in CMZ. Figure 8 shows the respective average estimated bond-specific safety premium series from these three major euro-area bond markets.

As already noted, the estimated bond-specific safety premia for German GBI's average -33 basis points over our sample period from October 2007 to December 2024, while the estimated safety premia of French OAT€s average -1 basis points for the shown period from October 2002 to December 2022. Hence, our results contrast with those reported by CM for French OAT€s in that the latter market contains no detectable safety premium on average. This sizable and persistent wedge in the pricing of bonds across these two otherwise seemingly similar markets implies that each market should be analyzed separately, just like nominal French OATs and German bunds are analyzed in isolation and not pooled. More importantly, we note that the estimated safety premia of German bunds average -124 basis points. Thus, the safety premia of nominal bunds are about four times larger than those estimated for the much smaller and less-liquid market for GBI's.

To contrast these results for the euro area with those reported in the literature for the United States, we note that U.S. TIPS prices contain a sizable liquidity premium discount,

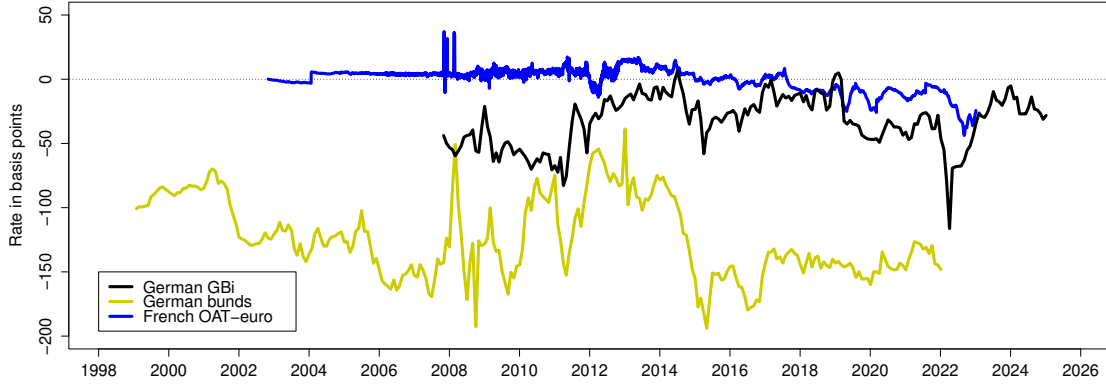


Figure 8: **Comparison of Average Estimated Bond-Specific Safety Premia**

Illustration of the average estimated bond-specific safety premium of German GBI bonds implied by the AFNS-R model estimated with a diagonal specification of $K^{\mathbb{P}}$ and Σ . Also shown are the average estimated bond-specific safety premium of standard German bunds reported by CMZ and the average estimated bond-specific safety premium of French OAT€s reported by CM.

which is well documented in the literature; see ACR, D’Amico et al. (2018), and Pflueger and Viceira (2016), among many others. Cardozo and Christensen (2026) offer a rationale for the illiquidity of inflation-linked bonds like TIPS. By being protected against inflation, indexed bonds are inherently less traded than nominal bonds. In addition, foreigners not exposed to the domestic price index do not benefit from owning them. Combined this significantly reduces their trading volumes, leading to a market for these bonds dominated by patient domestic buy-and-hold investors. This drives up the search frictions in the over-the-counter market for these bonds and leads to a steady-state outcome with their prices containing a large liquidity discount. Here, we find that, thanks to their high credit quality, GBI bonds are able to overcome this inherent illiquidity by offering euro-area investors a really safe asset to store their wealth.

4.2 Robustness Analysis

This section examines the robustness of the average safety premium reported in the previous section to some of the main assumptions imposed so far. Throughout the section, the AFNS-R model with diagonal $K^{\mathbb{P}}$ and Σ matrices serves as the benchmark model.

First, we assess whether the specification of the dynamics within the AFNS-R model matters for the estimated GBI bond-specific safety premium. To do so, we estimate the AFNS-R model with unconstrained dynamics, that is, the AFNS-R model with unrestricted $K^{\mathbb{P}}$ matrix and lower triangular Σ matrix. Figure 9 shows the estimated GBI bond-specific safety pre-

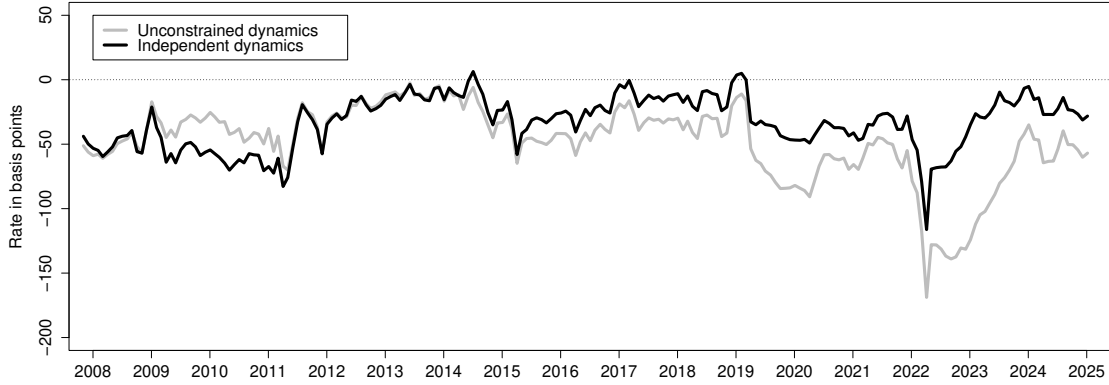


Figure 9: **Average Estimated GBi Bond-Specific Safety Premium: Alternative $\mathbb{P}$ Dynamics**

Illustration of the average estimated bond-specific safety premium of German GBi bonds for each observation date implied by the AFNS-R model when estimated with unconstrained dynamics as detailed in the text as well as independent factor dynamics with a diagonal specification of $K^{\mathbb{P}}$ and Σ . In both cases, the bond-specific safety premia are measured as the estimated yield difference between the fitted yield to maturity of individual GBi bonds and the corresponding frictionless yield to maturity with the bond-specific risk factor turned off.

mium from this estimation and compares it to the series produced by our benchmark model. Note that they are highly positively correlated. Thus, we conclude that the specification of the dynamics within the AFNS-R model only play a minor role for the estimated bond-specific safety premia, which is consistent with the findings of ACR in the context of U.S. TIPS.

Second, we assess whether the data frequency plays any role for our results. To do so, we estimate the AFNS-R model using daily, weekly, monthly, and quarterly data. Based on the results above, it suffices to focus on the most parsimonious AFNS-R model with diagonal $K^{\mathbb{P}}$ and Σ matrices. Figure 10 shows the average estimated GBi bond-specific safety premium series from all four estimations. In the ten-year period from 2012 to 2022 when there are consistently four or more GBi bonds outstanding, the state variables within the AFNS-R model are all well identified and not sensitive to the data frequency. As a consequence, the average estimated GBi bond-specific safety premia series are all very similar during this period. In contrast, prior to 2012 and towards the end of our sample when there are at most four or less GBi's trading, not all state variables in the model are fully identified. In that case, the estimated GBi bond-specific safety premia become sensitive to the shocks in the data, which vary with the data frequency. This explains the wider dispersion among the estimates in the early and later parts of our sample.

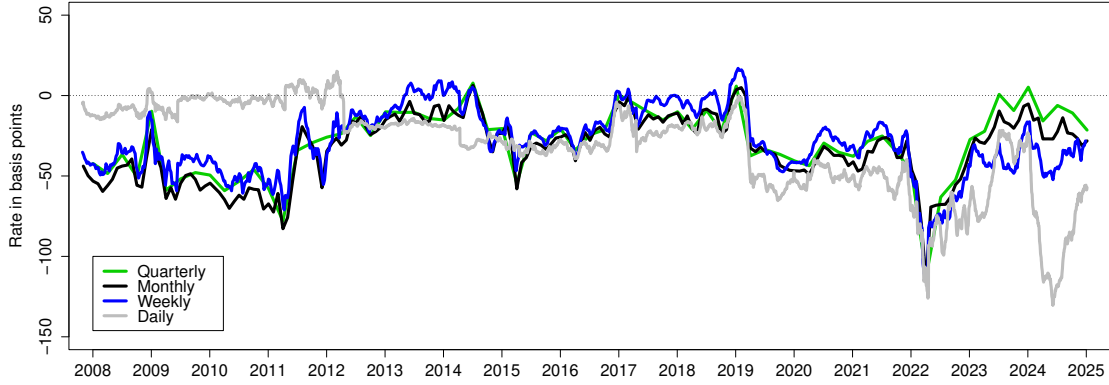


Figure 10: **Average Estimated GBI Bond-Specific Safety Premium: Data Frequency**

Illustration of the average estimated bond-specific safety premium of German GBI bonds for each observation date implied by the AFNS-R model with a diagonal specification of $K^{\mathbb{P}}$ and Σ when estimated using daily, weekly, monthly, and quarterly data. In all cases, the bond-specific safety premia are measured as the estimated yield difference between the fitted yield to maturity of individual GBI bonds and the corresponding frictionless yield to maturity with the bond-specific risk factor turned off.

4.3 Determinants of the GBI Safety Premium

In this section, we explore which factors matter for the size of the bond-specific safety premia in the GBI bond prices. To explain the variation of the average estimated safety premium series, we run regressions with it as the dependent variable and a wide set of explanatory variables that are thought to play a role for the bond-specific safety premia as explained in the following.

To begin, we are interested in the role of factors that are believed to matter for GBI market liquidity specifically or bond market liquidity more broadly as they could influence the estimated bond-specific safety premia. First, we include the average GBI bond age and the one-month realized volatility of the ten-year GBI bond yield as proxies for GBI bond liquidity following the work of Houweling et al. (2005).¹⁶ Inspired by the analysis of Hu et al. (2013), we also include a noise measure of German bund prices to control for variation in the amount of arbitrage capital available in the German government bond market.¹⁷ Furthermore, we

¹⁶The ten-year GBI bond yield is the ten-year fitted real yield implied by the estimated AFNS model using a sample that starts in June 2009, when the third GBI is issued.

¹⁷Given the small number of GBIs we cannot construct a noise measure for the GBI market. Instead, the noise measure used is the mean absolute fitted error from an estimated arbitrage-free generalized Nelson-Siegel (AFGNS) model of German bund prices; see Christensen et al. (2009). Note that each error is calculated as the difference between the observed bund price converted into yield to maturity and the fitted bund price also converted into yield to maturity. The data is described in Appendix A.

consider the ten-year yield spread of bonds issued by the German institute Kreditanstalt für Wiederaufbau (KfW) over those of matching German bunds. This is a widely used indicator of liquidity risk in the German government bond markets; see Monfort and Renne (2014).¹⁸ Finally, we add the euro overnight interbank rate to proxy for the opportunity cost of holding money and the associated liquidity premia of German government bonds, including GBi's, as explained in Nagel (2016). Combining these five explanatory variables tied to market liquidity and functioning produces the results reported in regression (1) in Table 5. We note a relatively modest adjusted R^2 of 0.36. The one-month realized volatility of the ten-year GBi yield and the noise measure both have statistically significant negative coefficients, while the average GBi bond age and the KfW-bund spread also have negative coefficients, although they are not significant. Overall, these results imply that an increase in the liquidity risk of GBi bonds is associated with *more negative* and hence larger average estimated bond-specific safety premia. Moreover, the overnight rate, which serves as a proxy for the opportunity cost of holding money, has a positive coefficient, which is inconsistent with the mechanisms detailed in Nagel (2016). Overall, we take these results to show that our average estimated bond-specific safety premia in the GBi prices do not behave like traditional liquidity premium discounts, which seems reasonable given that they are significantly negative on average and hence represent convenience safety premia.

After having explored the role of liquidity factors, we examine the effects of factors reflecting risk sentiment domestically and globally on the average estimated GBi bond-specific safety premia. This set of variables includes the VIX, which represents near-term uncertainty about the general stock market as reflected in options on the Standard & Poor's 500 stock price index and is widely used as a gauge of investor fear and risk aversion. The set also contains the yield difference between seasoned (off-the-run) U.S. Treasury securities and the most recently issued (on-the-run) U.S. Treasury security of the same ten-year maturity. This on-the-run (OTR) premium is a frequently used measure of financial frictions in the U.S. Treasury market. To control for factors related to the uncertainty about the interest rate environment, we include the MOVE index. The fourth variable is the U.S. TED spread, which is calculated as the difference between the three-month U.S. LIBOR and the three-month U.S. T-bill interest rate. This spread represents a measure of the perceived general credit risk in global financial markets. As an additional indicator of credit risk and credit risk sentiment across core euro-area government bond markets, we use the composite measure of the credit risk of French inflation-linked government bonds vis-à-vis German inflation-linked government bonds constructed by CM. The next variable in the set is the ten-year U.S. Treasury yield from the Federal Reserve's H.15 database, which is included to control for reach-for-yield effects in advanced economies. This may be particularly relevant for our sample during the period

¹⁸The construction of the KfW-bund yield spread is detailed in Appendix B.

Explanatory variables	(1)	(2)	(3)
Avg. bond age (yrs)	-2.501 (1.745)		2.415 (2.447)
One-month realized volatility of ten-year real yield (bps)	-0.792** (0.362)		-1.045*** (0.366)
Bund noise measure (bps)	-8.744*** (1.757)		-6.643*** (2.198)
Ten-year KfW-bund yield spread (bps)	-0.0917 (0.165)		-0.386* (0.207)
Overnight rate (%)	8.105*** (1.962)		5.274* (3.157)
VIX (%)		-1.050*** (0.288)	-0.462 (0.284)
Ten-year OTR premium (bps)		0.0374 (0.330)	0.849 (0.656)
MOVE Index (bps)		-0.215** (0.108)	-0.319** (0.134)
TED Spread (bps)		0.116*** (0.0414)	0.260*** (0.0968)
Composite credit risk measure (bps)		0.204*** (0.0631)	0.275*** (0.0915)
Ten-year US Treasury yield (%)		4.851 (3.690)	6.214 (4.026)
WTI (\$)		-0.246** (0.110)	-0.286* (0.146)
HICP inflation (%)		-0.509 (0.880)	2.832** (1.112)
German debt-to-GDP ratio (%)		-0.183 (0.510)	0.717 (0.650)
Constant	16.02 (10.59)	10.79 (37.13)	-40.10 (56.74)
Observations	186	204	183
Adj. R^2	0.362	0.361	0.529

Table 5: Regression Results for the Average Estimated GBi Bond-Specific Safety Premium

The table reports the results of regressions with the average estimated bond-specific safety premium of German GBi's as the dependent variable and 14 explanatory variables. Standard errors computed by the Newey-West estimator (with three lags) are reported in parentheses. Asterisks *, ** and *** indicate significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

between December 2008 and December 2015 and again in the 2020-2021 period when U.S. short-term interest rates were constrained by the zero lower bound. We also include the West Texas Intermediate (WTI) Cushing crude oil price to proxy for energy prices, which represent a significant risk to the inflation outlook in many countries around the world, including many euro area member states. Our final two variables represent factors that are fundamental to the prices of GBi's, namely HICP inflation and the German government debt-to-GDP ratio. The results of the regression with these nine explanatory variables are reported in regression (2) in Table 5. This produces a similar adjusted R^2 of 0.36. We note that five of the nine variables have explanatory power as their estimated coefficients are statistically significant.

To assess the robustness of the results from the first two regressions, we include all 14 variables with the results reported in column (3) in Table 5. This joint regression produces a notably higher adjusted R^2 of 0.53. The sizable increase in the adjusted R^2 suggests that there is only modest overlap between the two sets of explanatory variables. While the first set is squarely focused on the liquidity in the GBi market, the second set represents global risk sentiment and flight-to-safety effects along with two macro variables.

In the following, we elaborate on the interpretation of the estimated regression coefficients based on the results for the joint regression model reported in the last column of Table 5.

First, the negative coefficients on the ten-year GBi yield volatility, the bund noise measure, and the ten-year KfW-bund yield spread point to some flight-to-safety effects, whereby spells of heightened bond market liquidity risk seem to benefit the pricing of GBi's through lower and hence even more negative safety premia. The same interpretation applies to the negative coefficients on the VIX and the MOVE index, i.e., increased risk aversion in the U.S. stock market as captured by the VIX and increased uncertainty about the interest rate environment as reflected in the MOVE index both correlate with more negative safety premia in GBi bond yields, equivalent to higher bond prices.

In contrast, the positive coefficients on the TED spread and the composite credit risk measure indicate that perceptions about credit risk in financial markets more broadly as represented by the TED spread or in sovereign bond markets in the euro area more narrowly as captured by our composite credit risk measure both push the estimated GBi safety premium series higher. Based on these results we conclude that some part of the estimated bond-specific safety premia seems to reflect compensation for credit risk, but we leave it for future research to examine that conjecture further. Still, we take these findings to imply that systemic fears about the solvency of the global financial system or that of core euro-area governments as reflected in the TED spread and our composite credit risk measure, respectively, are associated with diminishing safety premia in GBi prices. These observations are also consistent with the findings of Christensen and Mirkov (2022), who document a lasting upward shift in the safety premia of Danish and Swiss government bonds following the introduction of the euro

in January 1999, meaning that these bonds became more valuable after the euro started circulating. The authors argue that the launch of the euro raised the prospect of potential scenarios with either a breakup of the euro area or some countries, presumably France or Germany, bailing out one or more other euro member states. Either way, such scenarios would bring into question the safe asset status of government bonds even in core euro-area countries and diminish their safety premia. The European sovereign debt crisis represents an example of this type of outcome as it involved an outright default by Greece along with material risks to the solvency of multiple other euro member states and even fears about the survival of the euro itself. Our results indicate that the safety premia of GBi bonds were greatly diminished during this period and its aftermath. In Figure 8, we also note a similar change to the safety premia of German bunds.

Importantly, during spells of non-systemic shocks that merely reflect elevated liquidity risks in financial markets or heightened risk aversion among global investors as captured by several of our explanatory variables,¹⁹ the GBi safety premia indeed behave like safety premia. We take these results to imply that GBi bonds serve a role as safe assets in the sense that investors are not (fire) selling these bonds during spells of financial market turmoil. Instead, they seem to value these bonds even more highly as their safety premia tend to increase under those circumstances. However, we are reluctant to refer to these effects as flight-to-safety effects because GBi's are hard to buy and sell on short notice, and the involved volumes are most likely modest in comparison with the trading volumes in the standard bund market.

Returning to the remaining variables, higher U.S. Treasury yields are likely to make euro-area safe assets less attractive in a relative sense all else being equal. This seems to account for the estimated positive coefficient on the U.S. Treasury yield with higher yields entailing an increase in our estimated safety premium series, meaning a reduction in the excess price GBi's can command in the market. However, we note that this effect is insignificant in both regressions (2) and (3).

Moreover, if energy prices increase as measured by the WTI oil price, inflation is likely to go up. In that case, inflation-indexed bonds become more convenient assets to hold. In our analysis, this shows up as a more negative safety premium and explains the negative coefficient on the WTI price series.

Finally, both higher HICP inflation and higher German debt-to-GDP ratio are associated with an increase in our estimated safety premium series, meaning a lower convenience premium of GBi's, but these effects appear less robust as both coefficients are negative and insignificant in regression (2).

With the systematic negative coefficients on the liquidity risk variables—and even on

¹⁹These variables include the GBi ten-year yield volatility, the bund noise measure, the KfW-bund yield spread, the VIX, and the MOVE index.

the VIX and the MOVE index—we feel that we can confidently reject the conjecture that our average estimated bond-specific risk premia in the GBi prices should represent liquidity premia. Hence, the trading dynamics in the GBi market seem to be fundamentally different from those prevailing in the U.S. TIPS market, where liquidity premium discounts are a well-documented phenomenon. Moreover, these findings are consistent with the results reported by CM for the French OAT€ market.

4.4 Effects of ECB Bond Purchases

In their paper, CMZ argue that central banks around the world have affected the supply of safe assets through their large-scale asset purchase programs, widely known as quantitative easing (QE), by buying quasi-safe bonds in exchange for truly safe reserves. They provide evidence of this transmission mechanism for the ECB’s bond purchases based on a panel of international safety premia, including those of German bunds. Given our interpretation of the GBi bond-specific risk premia as safety premia, they should be affected by the ECB bond purchases through this mechanism on par with the effects on the safety premia of German bunds. To test this hypothesis, we replicate the regression analysis of CMZ using our GBi safety premium series as the dependent variable. Notably, to obtain an interpretation of the results similar to theirs, we switch the sign of the estimated GBi safety premium so that it becomes mostly positive, and a larger positive value of the series corresponds to a larger safety premium paid by bond investors to store their wealth in these securities.

CMZ measure the average treatment effect of the ECB’s QE bond purchases on the safety premia using a regression equation that takes the form:

$$\bar{\Psi}_t = \alpha + \delta_{pspp} d_t^{pspp} + \delta'_c D_t + \sum_{l=0}^L \delta'_l X_{t-l} + \epsilon_t. \quad (7)$$

Here, d_t^{pspp} is the stock of bonds acquired by the ECB through its Public Sector Purchase Programme (PSPP), expressed as a percentage of nominal GDP in the euro area;²⁰ D_t and X_t are vectors of control dummies and continuous control variables, respectively; L is the number of lags included; and ϵ_t is a random residual. The estimate of δ_{pspp} measures the effect on the safety premia of a 1 percentage point change in the stock of bonds held by the ECB under the assumption that the confounding variables in the vector X_t are exogenous and that $E[\epsilon_t|X_t] = 0$.

We follow CMZ and control for a host of confounding factors. Specifically, we consider

²⁰In principle, this measure should only include the amount of quasi-safe assets that have been acquired by the ECB and replaced with safe reserves, i.e., bonds issued by high-risk countries in the periphery of the euro area, and should hence exclude safe bonds issued by highly rated countries in the core of the euro area. However, in the absence of that granular data, we follow CMZ and use the entire stock of purchased government-backed securities as a proxy for the amount of replaced quasi-safe assets.

	1	2
α	-178.753*** (42.127)	-103.173 (118.892)
δ_{pspp}	-0.396** (0.172)	-1.751** (0.666)
controls	Yes	Yes(6 lags)
No of obs	82	76
Adj. R^2	0.663	0.687
St. Err.	NW(4)	NW(4)

Table 6: Average Treatment Effects of ECB Asset Purchases on GBi bond safety premium

The table reports the estimated coefficients from running the regression in equation (7) with Newey-West standard errors (4 lags). The first column reports the regression with the simple set of control variables, while the second column contains the results including controls with L set to six lags. Note that the regression is only run for the period from January 2015 until December 2021, as in Christensen et al. (2025). Asterisks *, ** and *** indicate significance at the 10 percent, 5 percent and 1 percent levels, respectively.

the CBOE Volatility Index (VIX), the TED spread, the 10-year on-the-run premium in U.S. Treasuries, and the spread between the Italian and German 10-year government bond yield to proxy for investors' risk aversion, financial market uncertainty, and related demand for safe-haven assets;²¹ we use Germany's debt-to-GDP ratio to control for effects tied to the supply of government bonds;²² and the overnight interest rate in the euro area serves as a proxy for the opportunity cost of holding money and the associated liquidity premia of government bonds, as explained in Nagel (2016). We also include the average German bund age and the one-month realized volatility of the 10-year German bund yield as additional proxies for bond liquidity following the work of Houweling et al. (2005). Inspired by the analysis of Hu et al. (2013), we include a noise measure of German bund prices to control for variation in the amount of arbitrage capital available in the German government bond markets. We add the overnight federal funds rate to proxy for the U.S. safe-asset liquidity premium as in Nagel (2016), and we consider the MOVE volatility index to proxy for risk aversion in global bond markets. Finally, we add a dummy variable to indicate when there was a negative interest rate environment in the euro area.

We run the regression described in equation (7) both with no lags in the controls ($L = 0$) and with $L = 6$ lags of the control variables. We stress that the sample used in these regressions starts in January 2015 after the ECB had launched its first actual QE program and ends in December 2021 to focus squarely on the period with active bond purchases, as

²¹See Grisse and Nitschka (2015).

²²See Krishnamurthy and Vissing-Jorgensen (2012).

in CMZ. The results are reported in Table 6. We note that the coefficient on the ECB QE purchase variable is negative and statically significantly so in both regressions, in particular in the most conservative specification with six lags included. Moreover, the sizes of the estimated coefficients are similar to those reported by CMZ for the safety premia of German bunds. In terms of magnitudes, an increase in the ECB bond purchases equal to 1 percent of euro-area GDP will reduce the GBi safety premium by about 1.75 basis points based on the most conservative specification.

To summarize, we find a strong and statistically significant negative correlation between our estimated safety premium series for the GBi market and the ECB’s asset holdings as a share of nominal GDP in the euro area. Given that this measure of the ECB’s asset holdings represents a proxy for the amount of quasi-safe assets that has been replaced by safe central bank reserves, we take these results to show that the resulting increase in the supply of truly safe assets lowered the excess price that very safe government bonds in the euro area can command in financial markets. Hence, our findings point to an important transmission channel of QE that operates by altering the relative supplies of quasi-safe versus truly safe assets. This is a strong out-of-sample test of the mechanism described in CMZ that we hereby confirm. Moreover, we speculate that other measures of bond safety premia from the euro area or neighboring countries would be similarly affected. Finally, these results further support our interpretation of the GBi bond-specific risk premia as safety premia.

4.5 Impact on Breakeven Inflation

In this section, we demonstrate how breakeven inflation (BEI)—the difference between nominal and real yields at comparable maturities—is impacted by the presence of safety premia in the prices of GBi’s.

To illustrate the effect of accounting for the GBi safety premia on BEI rates, we first construct a standard measure of fitted BEI rates by deducting fitted real yields from a standard AFNS model estimated based on GBi prices from fitted nominal yields based on bund prices constructed as described in Appendix A. In Figure 11, this series is shown at the ten-year maturity with a solid black line.

Next, we construct the adjusted BEI rates that is cleansed of the GBi safety premia by deducting instead the frictionless real yield implied by the AFNS-R model estimated based on the GBi prices. This produces the adjusted BEI rate shown at the ten-year maturity with a solid gray line. Note that the adjusted BEI rate is systematically below the fitted BEI rate by 0.31 percent on average, which is an economically meaningful difference.

The key takeaway is that fitted BEI rates constructed from German bond prices should not be taken at face value. Instead, it is important to account for the safety premia before readings of their implied BEI rates are used to make judgements about investors’ inflation

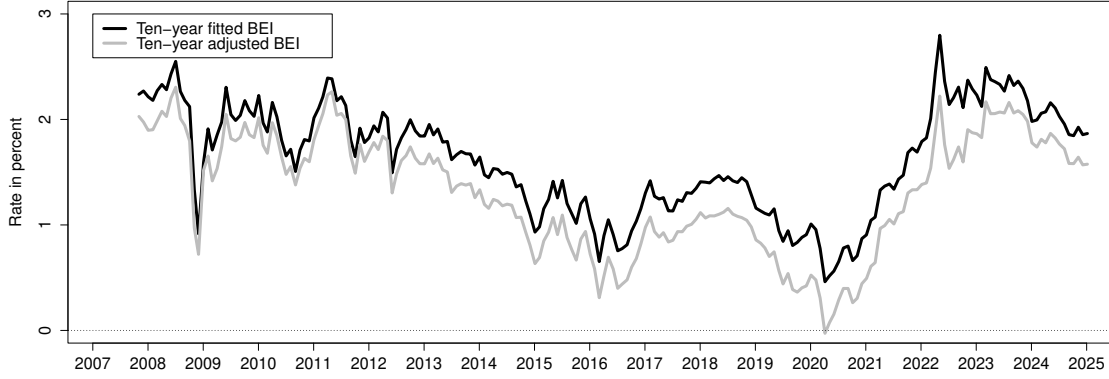


Figure 11: **Ten-Year BEI Rates**

Illustration of the ten-year fitted BEI rate obtained by fitting a generalized AFNS model to German bund prices as described in Appendix A and deducting the fitted real yields from a standard AFNS model estimated using our sample of GBi prices. Also shown is the ten-year adjusted BEI produced by taking the same fitted nominal bund yields as above and instead deduct the frictionless real yields implied by the AFNS-R model with a diagonal specification of $K^{\mathbb{P}}$ and Σ estimated using our sample of GBi prices. Both data series are monthly and cover the period from October 31, 2007, to December 30, 2024.

expectations or the premia they demand for assuming inflation risk.

Finally, while we demonstrate the impact of such an adjustment for the GBi safety premia, we stress that a complete assessment should bring in the safety premia of German bunds, but that is a task beyond the scope of this paper.

5 Termination of the GBi Program

In this section, we examine the market response to the German government’s decision to terminate the GBi program announced on November 22, 2023.

5.1 The Announcement

The decision to cease issuance of GBi bonds was made by the German Federal Finance Agency and communicated to the public through a simple press release on November 22, 2023. It contained the following two brief paragraphs:²³

“The Federal government has decided to withdraw from the market for inflation-linked

²³The press release is available at: https://www.deutsche-finanzagentur.de/fileadmin/user_upload/Pressemitteilung/en/2023/2023_11_22.PM.08.Federal_government_discontinues_programme_for_inflation-linked_bonds.pdf.

GBi	0.10% 4/15/2026	0.50% 4/15/2030	0.10% 4/15/2033	0.10% 4/15/2046
Maturity (yrs)	2.40	6.40	9.40	22.40
11/21-2023	110.7	39.2	37.0	43.5
11/22-2023	116.1	41.7	38.4	40.7
11/23-2023	112.5	34.1	30.9	28.8
1-day change	5.4	2.5	1.4	-2.8
2-day change	1.8	-5.1	-6.1	-14.7

Table 7: Response of German Inflation-Linked Government Bond Yields to GBi Termination Announcement

The table reports the one- and two-day responses of the outstanding German inflation-linked bond yields to the announcement terminating all future GBi issuance on November 22, 2023. All numbers are measured in basis points. The data are mid-market quoted yields to maturity at market close downloaded from Bloomberg.

bonds: From 2024, no further inflation-linked Federal securities will be issued, nor will already outstanding securities be reopened.

The currently outstanding inflation-linked Federal securities will continue to be tradable on the market. The remaining programme comprises four securities with a current total volume of €66.25 billion and remaining maturities between 2.5 to 22.5 years.”

The announcement caught investors and market observers by surprise as there had been no prior indication of any changes to the GBi program. Moreover, conversations with staff at the German Bundesbank indicate that Bundesbank was not consulted in this matter.

On its web site, the German Federal Finance Agency states:²⁴

“At €106 billion, the trading volume of inflation-linked Federal securities in 2023 was significantly lower than in the previous year – at €151 billion. Their share of the total trading volume of all Federal securities remained at the 2 percent recorded in 2022.”

5.2 The Market Reaction

In the following, we assess the impact of the termination of all future GBi issuance on both GBi yields directly and on our estimated GBi bond-specific risk premia.

Table 7 reports the one- and two-day yield changes for the GBi’s outstanding at the time of the announcement. The results indicate a notable market reaction with a two-day change in the observed GBi yield of almost -15 basis points at the 22-year maturity. Importantly, though, the yield declines on November 23, 2023, might reflect investors digesting the monetary policy accounts of the European Central Bank’s October 2023 governing council meeting

²⁴See <https://www.deutsche-finanzagentur.de/en/federal-securities/types-of-federal-securities/inflation-linked-federal-securities>.

that were released on November 23, 2023, where policymakers were cautiously optimistic about inflation falling in the euro zone.²⁵ Thus, we can only rely on the one-day responses as a guide to investors' reactions. These responses were all very timid. We interpret the yield increase for short- to medium-term GBI's as reflecting somewhat weaker expected liquidity going forward. In contrast, the yield decline for the single long-term GBI we take to reflect the positive impact of no competing supply coming to market for these long-term securities.

Since the responses in Table 7 reflect changes in yields to maturity, they are sensitive to both the bond coupon sizes and the shape of the underlying real yield curve and therefore hard to interpret and compare across bonds. For a cleaner read, Table 8 reports the responses of fitted nominal and real zero-coupon yields, where we focus on the important five- to ten-year maturity range that is commonly used in the construction of breakeven inflation measures.

As documented in Table 8, the two-day response is a firming in nominal yields of about 5 basis points and a decline in real yields of about 5 basis points. Thus, the net effect is a material firming in German bond-based BEI rates of about 10 basis points in the five- to ten-year maturity range. This could be interpreted as investors taking the announcement as a signal that the German government expects elevated inflation going forward. Without any other motivations provided for the termination of the GBI program, investors at a minimum would have to consider an economic rationale, namely that the government expects indexed debt to be expensive relative to nominal debt thanks to inflation remaining elevated above the ECB's 2 percent target.

To provide further support for this interpretation, we also examine the reaction of euro inflation swap rates, which are derivatives with cash flows tied to the change in HICP (ex tobacco), but not subject to any bond-specific premia unlike bunds and GBI's. The two-day changes in the five- to ten-year inflation swap rates are reported in the bottom panel of Table 8. Note that these rates also register a firming, although a notch lower at 3-4 basis points.

The uneven response between German bond-based BEI rates and euro inflation swap rates in the two-day window around the November 22, 2023 announcement is suggestive evidence that some part of the German bond price changes reflects a reaction to the termination of the GBI program in addition to a response to the news about the account of the October 2023 ECB governing council meeting as the latter in isolation should have produced a uniform reaction across the two markets. Hence, the market reaction provides some support for the notion that German bond investors took the sudden and unexpected termination of the GBI program as a sign that the German government saw the inflation outlook over the foreseeable future as less beneficial to inflation-linked debt issuance.

²⁵Based on the ECB website, the accounts cover the monetary policy meetings of the Governing Council, normally held every six weeks. They include a review of financial, economic and monetary developments and policy options as well as a summary of the discussions and decisions. The accounts are typically published four weeks after the meetings.

Nominal bund yields						
Maturity	5-year	6-year	7-year	8-year	9-year	10-year
11/21/2023	243.70	241.37	242.39	245.47	249.72	254.51
11/22/2023	245.11	242.22	242.74	245.38	249.24	253.67
11/23/2023	249.77	247.11	247.76	250.48	254.38	258.83
1-day change	1.41	0.85	0.35	-0.10	-0.49	-0.83
2-day change	6.07	5.74	5.37	5.00	4.65	4.33

Real GBi yields						
Maturity	5-year	6-year	7-year	8-year	9-year	10-year
11/21/2023	49.66	42.35	38.85	37.60	37.66	38.41
11/22/2023	51.76	43.96	40.03	38.44	38.20	38.70
11/23/2023	46.00	37.89	33.76	32.03	31.69	32.11
1-day change	2.10	1.60	1.18	0.84	0.54	0.29
2-day change	-3.66	-4.46	-5.08	-5.58	-5.97	-6.30

Euro inflation swap rates						
Maturity	5-year	6-year	7-year	8-year	9-year	10-year
11/21/2023	223.50	225.00	226.38	228.13	229.75	231.00
11/22/2023	222.00	223.70	225.15	227.13	228.75	230.25
11/23/2023	226.00	227.95	229.65	231.88	233.63	235.25
1-day change	-1.50	-1.30	-1.23	-1.00	-1.00	-0.75
2-day change	2.50	2.95	3.27	3.75	3.88	4.25

Table 8: **Response of German Bond Yields and Euro Inflation Swap Rates to GBi Termination Announcement**

The top panel reports the two-day response of German nominal government zero-coupon bond yields to the announcement by the German Federal Finance Agency to permanently terminate its issuance of GBi bonds on November 22, 2023. The middle panel reports the two-day response of German real government zero-coupon bond yields to the announcement. The bottom panel reports the two-day response of euro inflation swap rates to the announcement. All numbers are measured in basis points.

5.3 Impact on GBi Market Conditions

In this section, we explore how the decision affected the trading conditions and the functioning of the GBi market.

First, we examine bid-ask spreads of the outstanding set of GBi bonds. We think of these spreads as representative measures of the current trading conditions in the market for these bonds. Figure 4 shows four-week moving averages of the bid-ask spreads for each GBi. We first note the general upward trend in the bid-ask spread series caused by the fact that the bonds become more seasoned and less liquid as time passes. Importantly, though, there is no

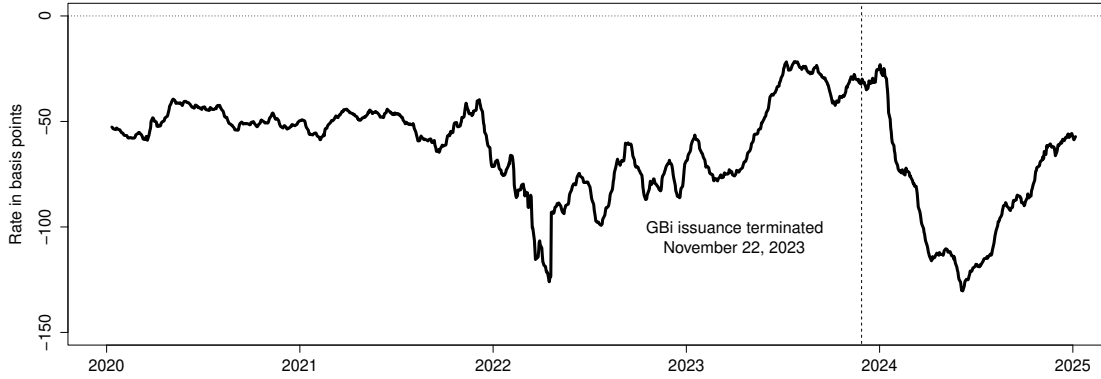


Figure 12: Average Daily Estimated Bond-Specific Safety Premium since 2020

Illustration of the average estimated GBi bond-specific safety premium for each observation date implied by the AFNS-R model estimated at daily frequency with a diagonal specification of $K^{\mathbb{P}}$ and Σ . The GBi bond-specific safety premia are measured as the estimated yield difference between the fitted yield to maturity of individual GBi bonds and the corresponding frictionless yield to maturity with the bond-specific risk factor turned off. The shown data cover the period from January 2, 2020, to December 30, 2024.

major change in the general bid-ask spread levels in the period following the November 22, 2023, announcement. Thus, the GBi trading conditions do not seem to have fundamentally changed.

Second, we assess whether there seems to have been any impact on the performance of the AFNS-R model and its ability to fit the GBi bond prices. To that end, we estimate the model using daily data instead of the monthly frequency considered so far. Figure 12 shows the resulting average estimated GBi bond-specific safety premium series since the start of 2020 through the end of December 2024. Importantly, there is hardly any reaction in the average estimated GBi bond-specific safety premium in the days following the announcement to cease all future GBi issuance. Thus, based on our model results, investors did not seem to worry much about the future liquidity in the GBi market despite no new supply being issued. Interestingly, though, the average estimated GBi bond-specific safety premium was clearly well below its historical average throughout 2022 and well into 2023, meaning that GBi's were trading at particularly high prices during this period that preceded the decision to cease future issuance. Normally, this would be the market signal to *increase* issuance.

As for the model fit specifically, Figure 5 shows the monthly fitted error series for all nine GBi bonds in our sample going back to October 2007. We note that there is no discernible tendency for larger or more volatile fitted errors since late 2023. Thus, the AFNS-R model has clearly maintained its ability to fit the cross section of GBi prices.

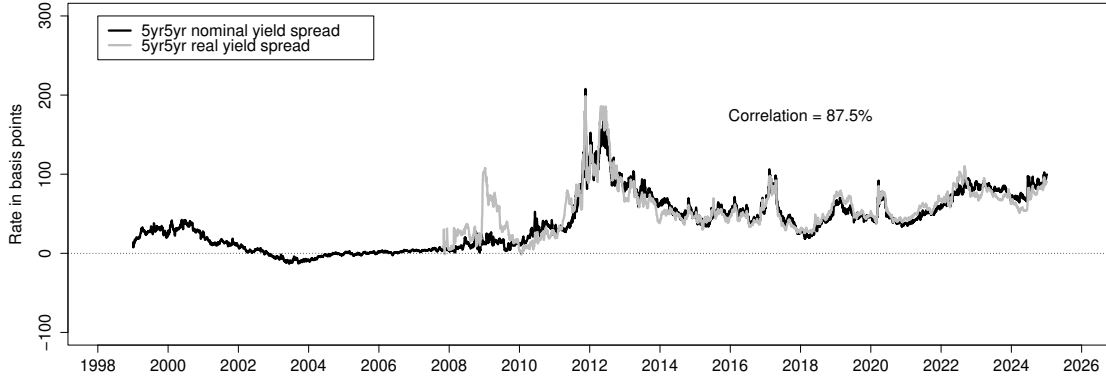


Figure 13: **Franco-German Government Bond Yield Spreads**

In related research, Christensen, Rudebusch, and Shultz (2025) analyze the announcement by the Canadian Finance Department on November 2, 2022, to permanently cease issuance of its inflation-linked bonds, known as Real Return Bonds (RRB). They also find a very modest response similar to our results above.

To go beyond the analysis so far and that of Christensen, Rudebusch, and Shultz (2025), we gauge to what extent GBI's are priced differently than standard German bunds. To do so, we contrast the pricing of each of these two classes of bonds with their respective French counterpart. If bunds are priced fundamentally differently than GBI's, which could be the case based on the notable difference between the average safety premia in the two markets documented in Figure 8, it should show up as differences in the yield spreads relative to their French counterparts. Specifically, as a representative measure of such spreads, we consider the five-year forward yield spread between French and German inflation-linked bond yields for a period starting five years ahead. Due to the late launch of the German inflation-linked government bond program, we can only construct this 5yr5yr Franco-German real yield spread starting in October 26, 2007. The available series since then through the end of our sample is shown with a solid gray line in Figure 13. We follow CM and interpret the 5yr5yr Franco-German real yield spread as mainly reflecting differences in credit risk premia rather than differences in liquidity risk premia.

We then calculate the matching 5yr5yr yield spread between standard French nominal government bonds, known as OATs, and German bunds, which is available back to January 1999 and shown with a solid black line in Figure 13.²⁶ Given the high liquidity of German bunds and their associated flight-to-safety status during spells of elevated financial market

²⁶See Appendices A and C for details of the data and the construction of these yields.

illiquidity, we acknowledge that this spread likely reflects elements of both the liquidity and safety premium advantages of the bund market relative to the French market for OATs. Against that background, the similarity between the 5yr5yr nominal and real Franco-German bond yield spreads is both evident and striking. For the overlapping period since October 26, 2007, with a total of 4,459 daily observations, the correlation in levels is an astonishing 87.5 percent, while the correlation in first differences at daily frequency is 63.2 percent. Equally importantly, the spreads are of nearly identical magnitudes, with means of 54.5 basis points and 56.5 basis points, respectively, for the overlapping period, while their corresponding standard deviations are 29.7 basis points and 28.9 basis points, respectively. The visual and statistical similarity implies that these two spread series are statistically indistinguishable from each other.

We find that the spreads do show spikes during flight-to-safety episodes: The Global Financial Crisis, the European sovereign debt crisis, the first Trump presidential victory in November 2016, and the peak of the COVID-19 pandemic, to name a few. However, these are all short-lived in nature, and the yield spreads do not revert to zero afterwards, rather they remain elevated with meaningful variation even in normal times. These observations combined lead us to conclude that the spreads reflect safety premia. Based on this interpretation, we also note that the safety of German government bonds has increased relative to that of French government bonds since 2018 as both spread series have trended higher in tandem.

There is an equally important takeaway, namely that GBi's appear to be as beneficial to issue when benchmarked against their French inflation-linked equivalents as regular nominal German bunds vis-à-vis their French nominal OAT equivalents. The yield spread saved is statistically indistinguishable with few exceptions for the available seventeen-year period. As a consequence, it is nearly impossible to pinpoint a time when it would have made a material difference to issue only nominal bunds as opposed to the mix of bunds and GBi's that were actually issued during our sample period. If anything, in the counterfactual with greater bund issuance and fewer GBi's outstanding, the spreads might have been favorable towards greater GBi issuance due to their increased scarcity.

To summarize, we find that neither the safety premia nor the trading conditions of the inflation-linked bonds were negatively impacted by the decision to cease future issuance of GBi bonds. Hence, for now, inflation-linked trading remains active even though no new issuance has come to market since November 2023. However, as time goes by and the few remaining GBi's start to reach maturity, the available number of bonds will inevitably shrink. This will exacerbate the problems with weak identification of the state variables within the AFNS-R model identified in Section 4.2. Over time, this will cause the model results to be overall less robust and reliable. Hence, although the GBi market appears to be well functioning at this point, the eventual shrinking market for these bonds is a cause for concern and makes

us caution against relying too heavily on this market in the longer run. However, obviously, for historical analysis of key events in the euro area during our sample period, this market is and will remain a rich and useful data source.

6 Conclusion

In this paper, we provide an in-depth analysis of the little-known German market for inflation-linked government bonds. As the first to assess this market, we document the existence of large convenience premia in the prices of these bonds that average 0.33 percent over our sample, meaning that investors are willing to overpay to own these bonds. Given their relatively low liquidity and related lack of moneyiness, we refer to these convenience premia as safety premia. Regression analysis with a large battery of explanatory variables supports this interpretation.

Even though these bonds were highly priced, the German Federal Finance Agency decided in November 2023 to cease all future issuance of such bonds as well as any reopening of existing ones. We examine the market reaction to this consequential announcement and find that neither the trading of these bonds nor their safety premia have been negatively affected by the fact that no new supply will come to market going forward. Hence, though this overlooked and understudied market provides a rich source of information about real rates in the euro area, we caution against putting too much weight on this market in the long run.

For now, though, we see the German bond market as a promising candidate for the construction of breakeven inflation for the euro area that can be decomposed into investors' expected inflation and the residual inflation risk premium. However, to succeed in such an endeavor, we stress that it is imperative to account for the safety premia in GBI prices documented in this paper as well as those in German bund prices documented in CMZ. This is a demanding task we leave for future research.

A Appendix: German Nominal Government Bond Data

In this appendix, we describe the data for the German nominal government bonds we use to construct some of the explanatory variables appearing in the regression analysis in Section 4.3 and the German nominal yield series analyzed in Section 5.3. The sample contains standard fixed-coupon bonds denominated in euros and commonly referred to as bunds.

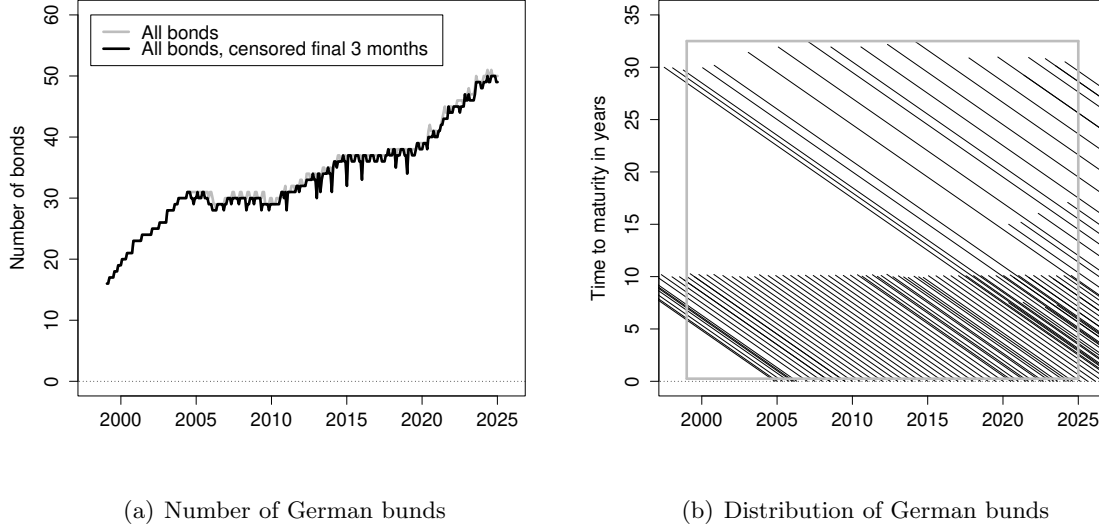


Figure A.1: **Overview of the German Nominal Government Bond Data**

Panel (a) reports the number of outstanding German nominal government bonds at a given point in time. Panel (b) shows the maturity distribution of German bunds in the sample. The solid gray rectangle indicates the sample used in our analysis, where the sample is restricted to start on January 31, 1999, and limited to bund prices with more than three months to maturity after issuance.

The number of outstanding German bunds over time in our sample is shown with a solid gray line in Figure 1(a). At the end of our sample period—which runs from January 1999 to December 2024—50 bonds were outstanding. However, as is widely recognized, prices of bonds near their maturity tend to be somewhat erratic. Therefore, to facilitate model estimation, we drop bonds from our sample when they have less than three months to maturity. Using this cutoff, the number of bonds in the sample is modestly reduced as shown with a solid black line in Figure 1(a).

Generally, the German Federal Finance Agency has issued a limited set of standard fixed-coupon bonds focusing mostly on ten- and thirty-year bonds mixed with a limited number of seven- and fifteen-year bonds during our sample period. The maturity distribution of the 100 bonds in our sample is shown in Figure 1(b). Each bond is represented by a single downward-sloping line that plots its remaining years to maturity for each date. As a consequence of the stable issuance pattern, there is little variation in the maturity range covered by the

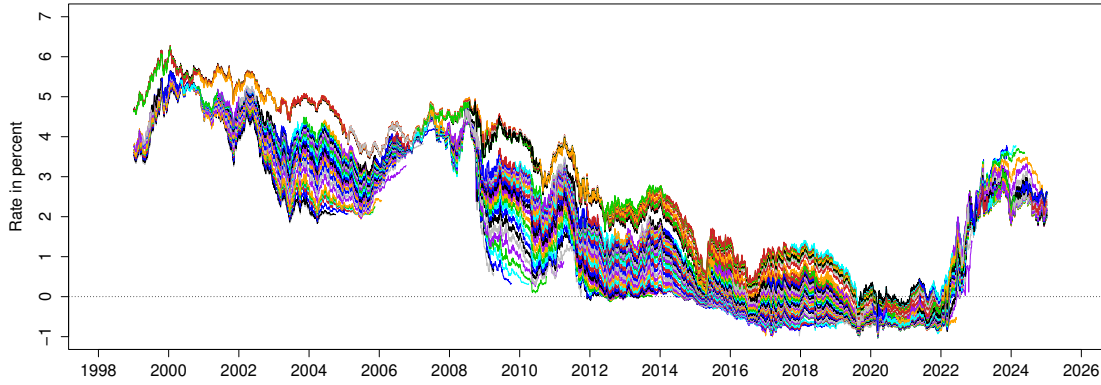


Figure A.2: Yield to Maturity of German Nominal Government Bonds

Illustration of the yield to maturity implied by the German bund prices considered in this paper, which are subject to two sample choices: (1) sample limited to the period from January 4, 1999, to December 30, 2024; (2) censoring of a bond's price when it has less than three months to maturity. Each bond yield series is shown with its own colored line.

outstanding bonds in our sample. This ensures the identification of the state variables in our term structure model throughout the sample period.

Figure A.2 shows the German bund prices converted into yield to maturity. Several things are worth noting regarding these yield series. First, there is a trend lower in the general yield level during this period from roughly 5 percent in the early 2000s to below zero by 2021 followed by a sharp reversal during the remaining years of our sample. Second, there is pronounced business cycle variation in the shape of the yield curve around the lower trend. The yield curve tends to flatten ahead of recessions and steepen during the initial phase of economic recoveries.

To construct fitted zero-coupon yields for this data set, we use the very flexible arbitrage-free generalized Nelson-Siegel (AFGNS) model developed in Christensen et al. (2009). With its five state variables, this model produces a very tight fit to the entire term structure of bond prices. As a consequence, our fitted German nominal zero-coupon bond yields can be considered to be of the highest possible quality as we are using the entire available universe of bund prices.

B Appendix: KfW Bond Data

In this appendix, we describe the data for the bonds issued by the German institute Kreditanstalt für Wiederaufbau (KfW) that we use to construct the ten-year KfW-bund yield spread used as an explanatory variable in the regression analysis in Section 4.3. The sample contains standard fixed-coupon bonds denominated in euros with pricing information available for at least half of the business days during the life of each bond. Finally, bonds with unreasonable and erratic price patterns were dropped. We consider the resulting sample of 109 bonds to be representative of the market for KfW bonds during our sample period.²⁷

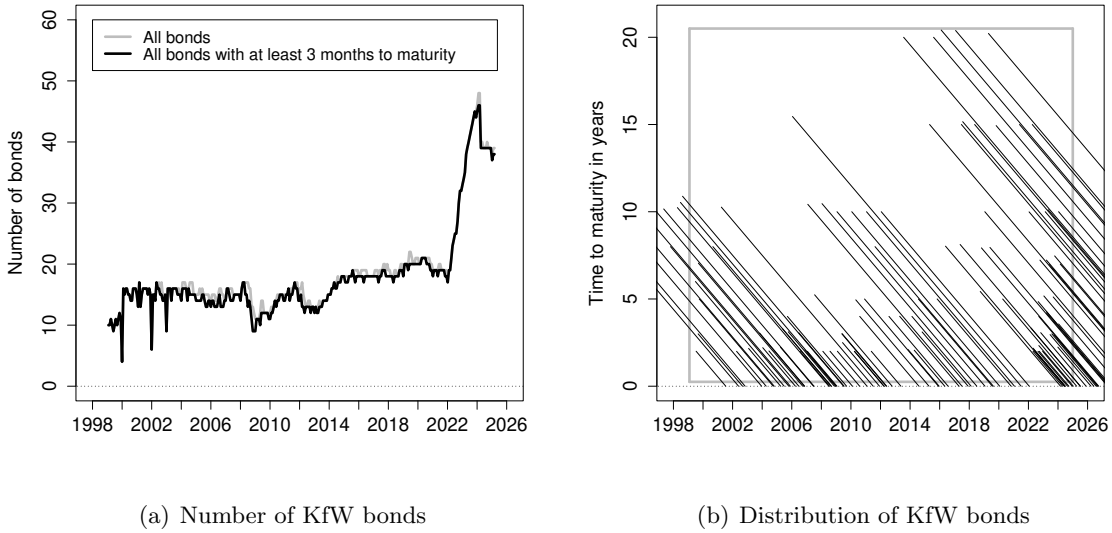


Figure B.3: **Overview of the KfW Bond Data**

Panel (a) reports the number of outstanding KfW bonds at a given point in time. Panel (b) shows the maturity distribution of the KfW bonds in the sample. The solid gray rectangle indicates the sample used in our analysis, where the sample is restricted to start on January 31, 1999, and limited to KfW bond prices with more than three months to maturity after issuance.

The number of outstanding KfW bonds over time in our sample is shown with a solid gray line in Figure 3(a). At the end of our sample period—which runs from January 1999 to December 2024—39 bonds were outstanding. However, as already noted in Appendix A, prices of bonds near their maturity tend to be somewhat erratic. Therefore, to facilitate model estimation, we drop bonds from our sample when they have less than three months to maturity. Using this cutoff, the number of bonds in the sample is modestly reduced as shown with a solid black line in Figure 3(a).

Generally, the KfW has issued a variety of standard fixed-coupon bonds with original ma-

²⁷For example, our sample includes eight of the nine KfW bonds mentioned explicitly in Monfort and Renne (2014). The missing KfW bond (4.25% 7/4/2014) had no price information on Bloomberg.

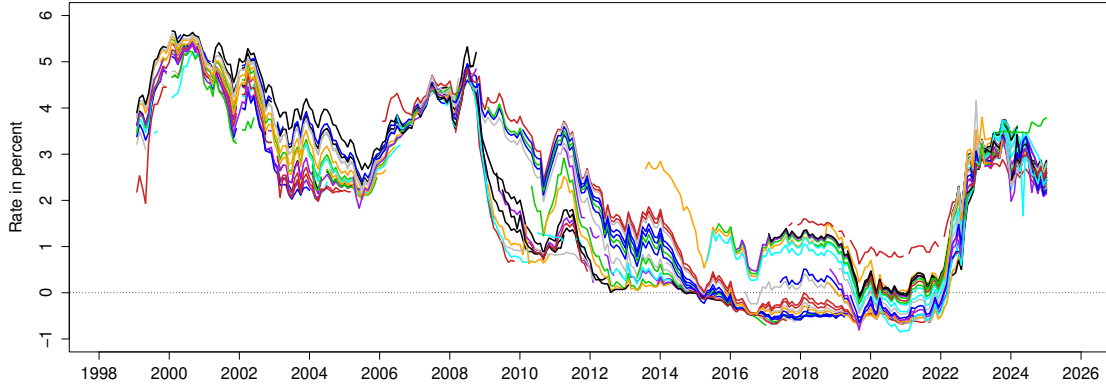


Figure B.4: **Yield to Maturity of KfW Bonds**

turities ranging from 2 years to 20 years during our sample period. The maturity distribution of the 109 bonds in our sample is shown in Figure 3(b). Each bond is represented by a single downward-sloping line that plots its remaining years to maturity for each date. Overall, there is significant variation in the maturity range covered by the outstanding bonds in our sample. However, our estimation method that uses all information in the entire panel of bond prices is well suited to handle this.²⁸

Figure B.4 shows the KfW bond prices converted into yield to maturity. Several things are worth noting regarding these yield series. First, there is a lower trend in the general yield level during this period from roughly 5 percent in the early 2000s to around zero by the end of our sample. Second, there is pronounced business cycle variation in the shape of the yield curve around the lower trend. The yield curve tends to flatten ahead of recessions and steepen during the initial phase of economic recoveries. These characteristics are the practical motivation behind our choice of using a three-factor model for the KfW yield curve, adopting an approach similar to what is standard for U.K. and U.S. data; see Christensen and Rudebusch (2012).

To construct the spreads between the yields of KfW bonds and those of German bunds, we first obtain fitted German bund zero-coupon yields by estimating the arbitrage-free generalized Nelson-Siegel (AFGNS) model developed in Christensen et al. (2009) using our sample of German bund prices. We then obtain fitted KfW zero-coupon yields by estimating the simpler arbitrage-free Nelson-Siegel (AFNS) model described in Christensen et al. (2011) using the sample of KfW bond prices described above. By deducting the former from the latter at the fixed ten-year maturity, we get an estimate of the ten-year yield spread shown in Figure

²⁸Finlay and Wende (2012) examine prices from a limited number of Australian inflation-indexed bonds using the extended Kalman filter for estimation similar to our approach.

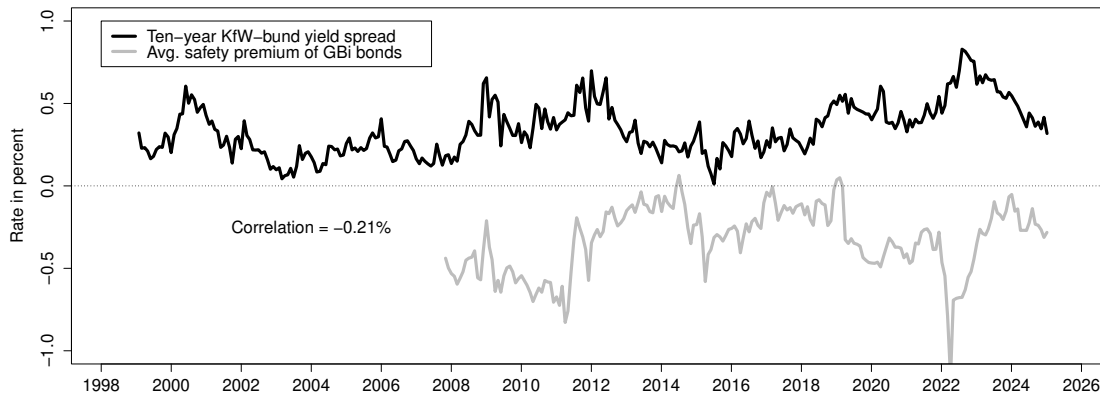


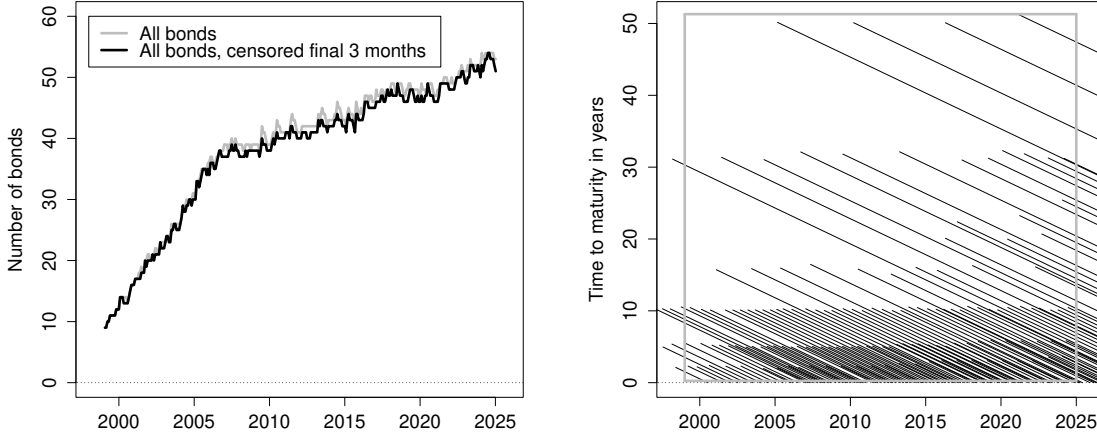
Figure B.5: Ten-Year KfW-Bund Yield Spread and Average GBi Bond Safety Premium

B.5.

CMZ argue that KfW-bund yield spreads should contain a minimum of safety premia as they cancel out in the yield spread calculation. Instead, KfW-bund yield spreads mainly represent a measure of the relative liquidity risk premia across the two bond markets. In contrast, our estimated bond-specific risk premia likely reflect both liquidity risk discounts and safety price premia. Importantly, we only observe the net effect, which is a large price premium that makes us refer to them as safety premia. This also explains the negative correlation between the two series reported in Figure B.5.

C Appendix: French Nominal Government Bond Data

In this appendix, we describe the data for the French nominal government bonds we use in the empirical analysis in Section 5.3. The sample contains standard fixed-coupon bonds denominated in euros and known as OATs.²⁹



(a) Number of French nominal government bonds (b) Distribution of French nominal government bonds

Figure C.6: Overview of the French Nominal Government Bond Data

Panel (a) reports the number of outstanding OATs at a given point in time. Panel (b) shows the maturity distribution of the OATs in the sample. The solid gray rectangle indicates the sample used in our analysis, where the sample is restricted to start on January 31, 1999, and limited to OAT prices with more than three months to maturity after issuance.

The number of outstanding OATs over time in our sample is shown with a solid gray line in Figure 6(a). At the end of our sample period—which runs from January 1999 to December 2024—53 bonds were outstanding. However, as is widely recognized, prices of bonds near their maturity tend to be somewhat erratic. Therefore, to facilitate model estimation, we drop bonds from our sample when they have less than three months to maturity. Using this cutoff, the number of bonds in the sample is modestly reduced as shown with a solid black line in Figure 6(a).

Generally, the French government has issued a variety of standard fixed-coupon bonds with original maturities ranging from 2 years to 50 years during our sample period. The maturity distribution of the 150 bonds in our sample is shown in Figure 6(b). Each bond is represented by a single downward-sloping line that plots its remaining years to maturity for each date. Overall, there is little variation in the maturity range covered by the outstanding bonds in our sample. As in the German bund data, this ensures the econometric identification

²⁹This represents an update of the sample analyzed in Christensen et al. (2024).

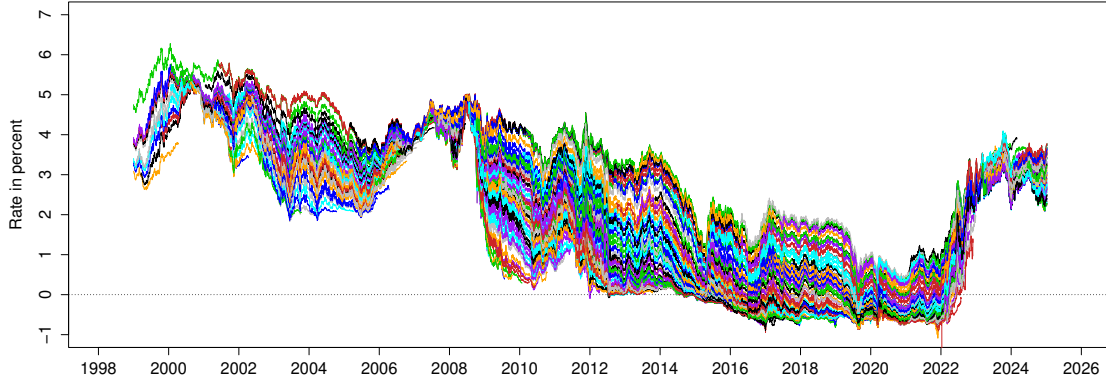


Figure C.7: Yield to Maturity of French Nominal Government Bonds

Illustration of the yield to maturity implied by the French government bond prices considered in this paper, which are subject to two sample choices: (1) sample limited to the period from January 4, 1999, to December 30, 2024; (2) censoring of a bond's price when it has less than three months to maturity. Each bond yield series is shown with its own colored line.

of the state variables in our model throughout our sample.

Figure C.7 shows the OAT bond prices converted into yield to maturity. Several things are worth noting regarding these yield series. First, there is a trend lower in the general yield level during this period from roughly 5 percent in the early 2000s to below zero by 2021 followed by a sharp reversal during the remaining years of our sample. Second, there is pronounced business cycle variation in the shape of the yield curve around the lower trend. The yield curve tends to flatten ahead of recessions and steepen during the initial phase of economic recoveries.

To construct fitted zero-coupon yields for this data set, we follow Christensen et al. (2024) and estimate the arbitrage-free generalized Nelson-Siegel (AFGNS) model developed in Christensen et al. (2009). With its five state variables, this model produces a very tight fit to the entire term structure of bond prices, as also demonstrated by Christensen et al. (2024). As a consequence, our fitted French nominal zero-coupon bond yields can be considered to be of the highest possible quality as we are using the entire available universe of OAT prices.

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